

4. Electricity & Magnetism

YOUR NOTES



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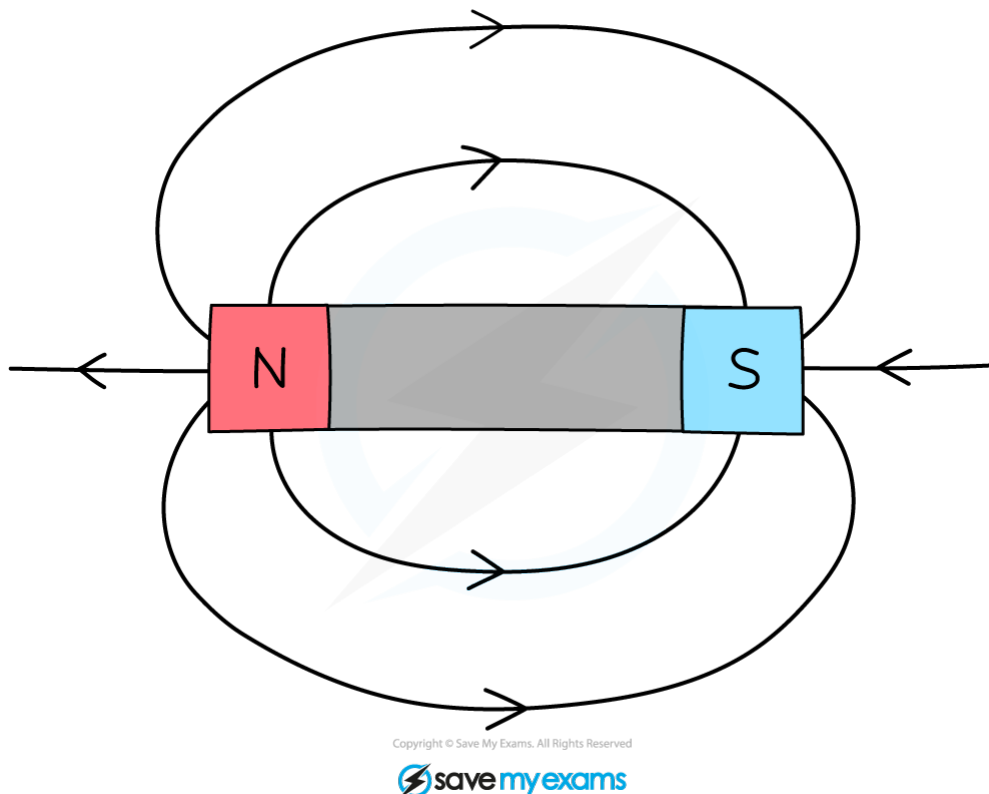
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4.1 SIMPLE PHENOMENA OF MAGNETISM

4.1.1 MAGNETISM

Magnetic Fields

- In the space around a magnet there is a **magnetic field**
- **Magnetic field lines** are a useful way of helping us to picture that field: Its strength and its direction



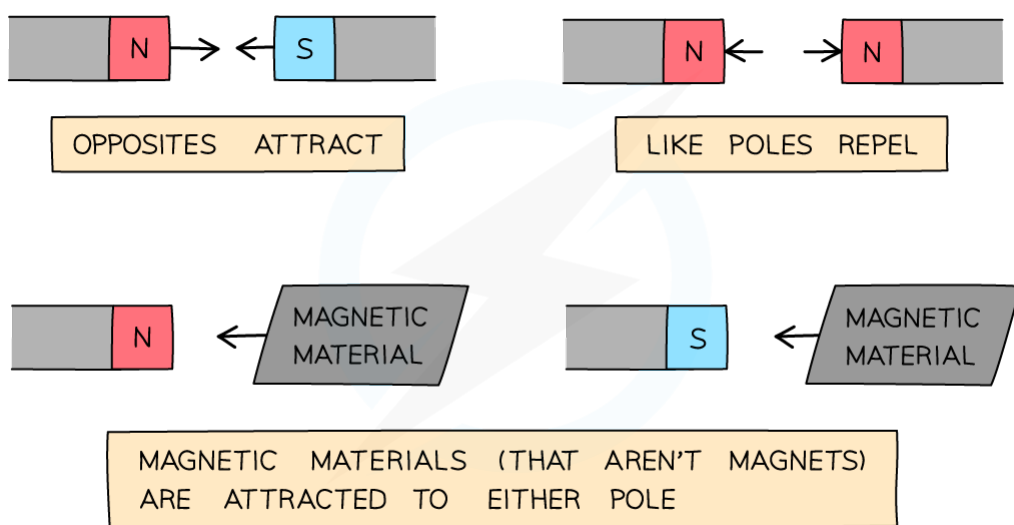
The magnetic field around a bar magnet

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- Magnetic field lines obey a couple of rules:
 - They always go from north to south ((indicated by an arrow midway along the line))
 - They never touch or cross other field lines
- When two magnets are held close together, there will be a force between the magnets:



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Opposite poles attract; like poles repel

- Magnetic materials (which are not magnets) will always be attracted to the magnet, regardless of which pole is held close to it
- **A magnet can only repel another magnet**
(This can be a useful test for a magnet)

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Magnetic Forces

- Magnetic forces are caused by the interaction of magnetic fields

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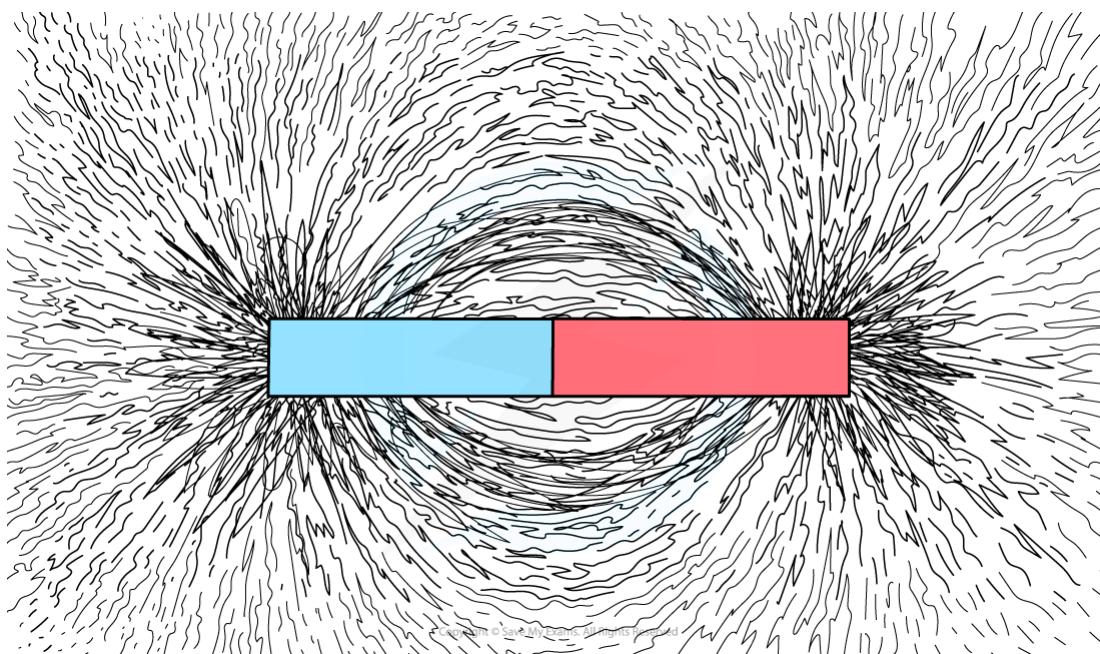


Plotting a Magnetic Field

- In your IGCSE examination you might be asked to describe a method of plotting the magnetic field around a bar magnet
- There are two principle ways of doing this:

Using Iron Filings

- Place a piece of paper on top of the magnet
- Gently sprinkle iron filings on top of the paper
- Now carefully tap the paper to allow the iron filings to settle on the field lines



Iron filings can be used to plot a magnetic field

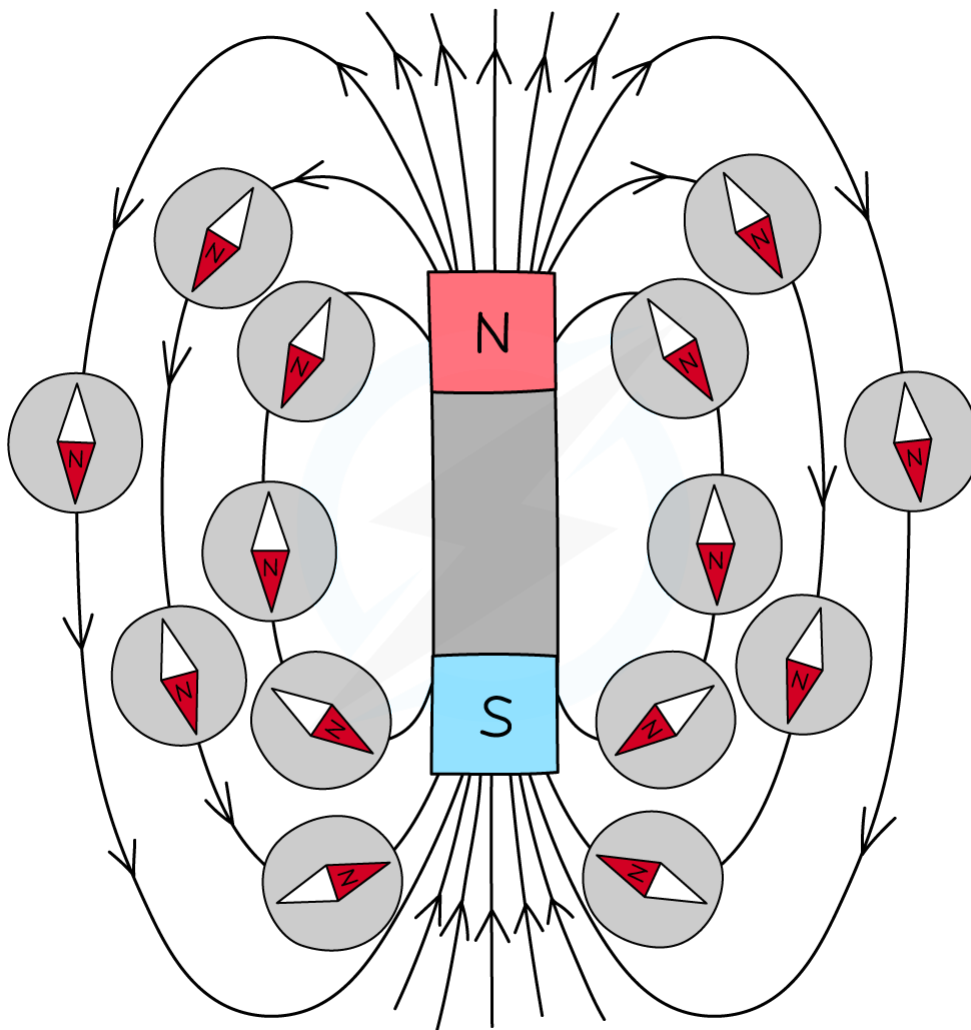
- Place the magnet on top of a piece of paper
- Draw a dot at one end of the magnet (near its corner)
- Place a plotting compass next to the dot, so that one end of the needle of the compass points towards the dot
- Use a pencil to draw a new dot at the other side of the compass needle
- Now move the compass so that it points towards the new dot, and repeat the above process

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- Keep repeating until you have a chain of dots going from one end of the magnet to the other. Then remove the compass, and link the dots using a smooth curve - the magnetic field line
- The direction of the field line is the same as the direction of the plotting compass
- You can now repeat the whole process several times to create several other magnetic field lines

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Compasses can be used to plot the magnetic field around a bar magnet

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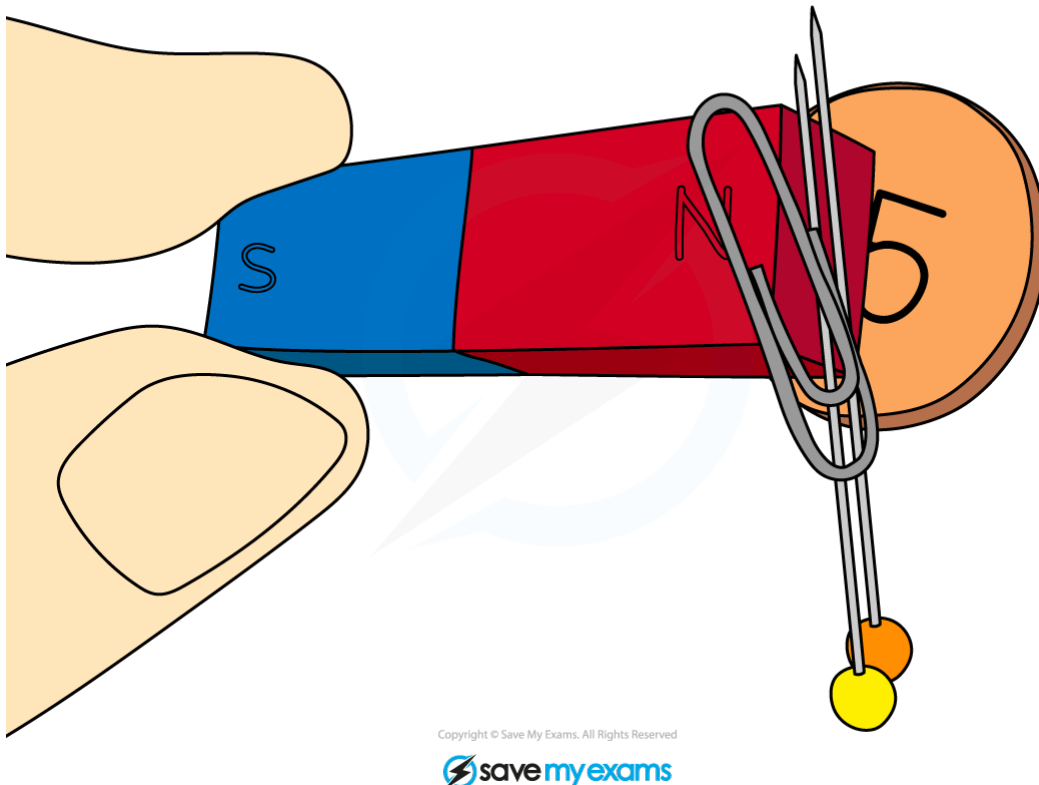
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4.1.2 MAGNETIC MATERIALS

What are Magnetic Materials?

- A magnetic material is a material that experiences a force when placed in a magnetic field
- Although all magnetic materials are metallic, not all metals are magnetic
- **Common magnetic materials include:**
 - Iron
 - Steel (an alloy of iron)
 - Nickel
 - Cobalt
- Note: **Copper and Aluminium are non-magnetic**

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Magnetic materials, such as iron, steel, nickel (the coin) and cobalt, are attracted to magnets

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Hard and Soft Materials

- **Magnetically soft materials** (e.g. Iron):
 - Are easy to magnetise
 - Easily lose their magnetism
- **Magnetically hard materials** (e.g. Steel):
 - Are hard to magnetise
 - Do not easily lose their magnetism
- Permanent magnets are made out of magnetically hard materials, as we don't want them to lose their magnetism
- Electromagnets are made out of magnetically soft materials, as we want them to be able to easily gain and lose their magnetism

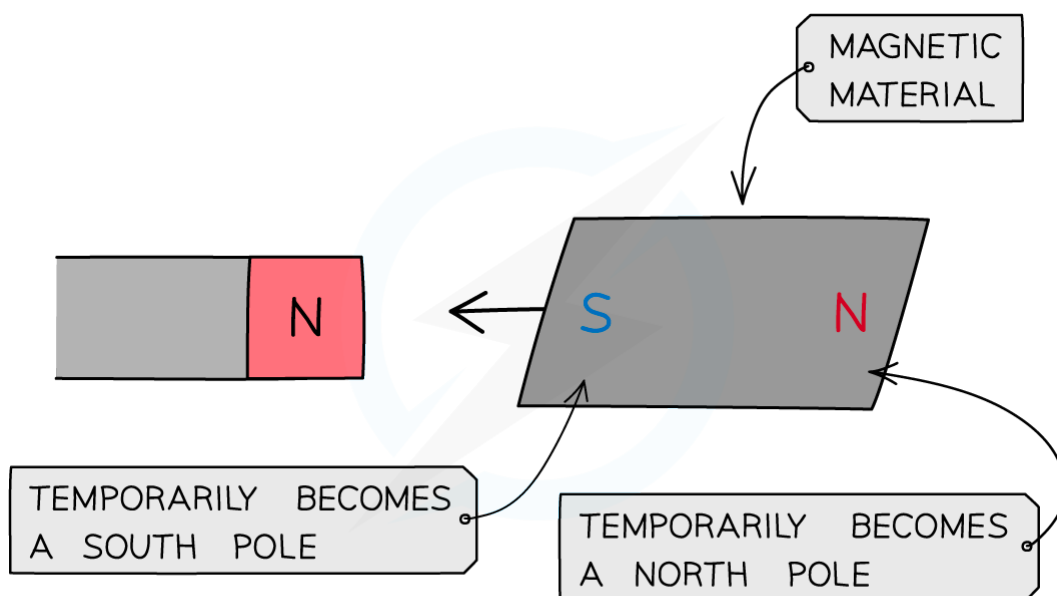
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Induced Magnetism

- When a magnetic material is placed in a magnetic field, the material can temporarily become magnetised:
 - One end of the material will become a north pole
 - The other end will become a south pole

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When a magnet is held close to a magnetic material, the material temporarily becomes magnetised

- This process is known as **magnetic induction** and results in a force being exerted on the material
- When the material is removed from the magnetic field, it will start to lose its magnetism (depending on whether it is magnetically hard or soft)

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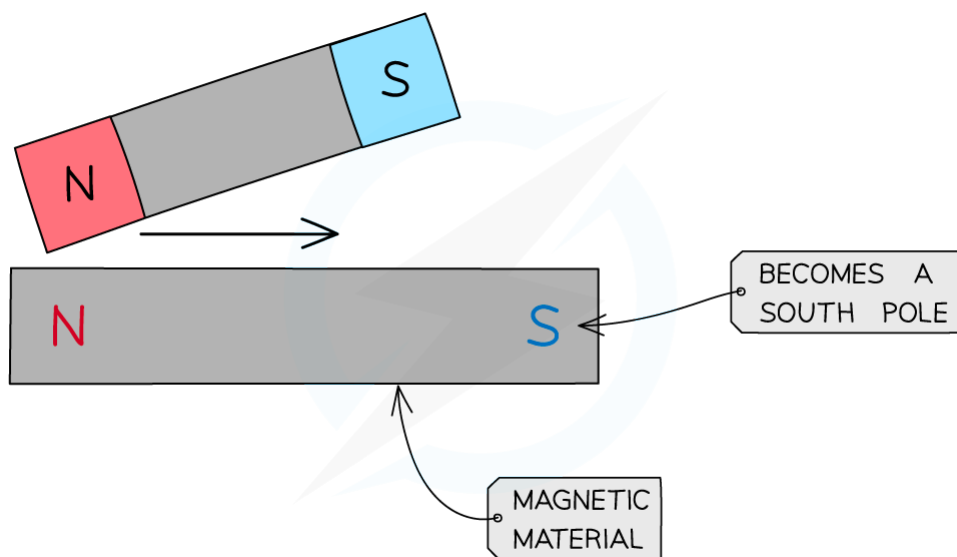


4.1.3 MAGNETS & ELECTROMAGNETS

Creating and Destroying Magnets

- Magnets are usually made from hard magnet materials
- There are several ways in which a magnet can be created:

Stroking with a magnet:



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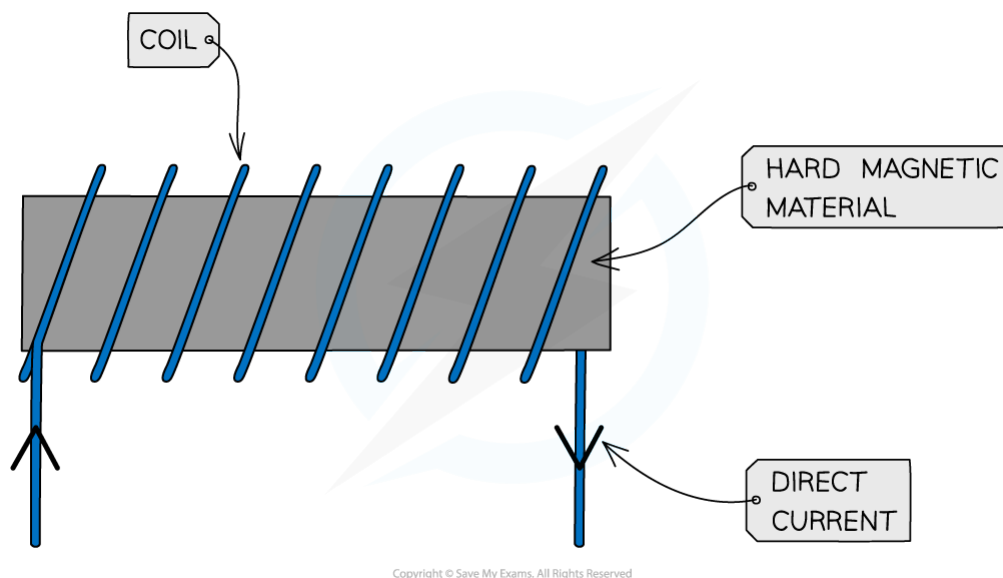
A magnetic material can be magnetised by stroking it with one end of another magnet

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Using a direct current (d.c.) in a coil:



Placing a magnetic material in a coil and then passing a direct current through the coil will also magnetise it

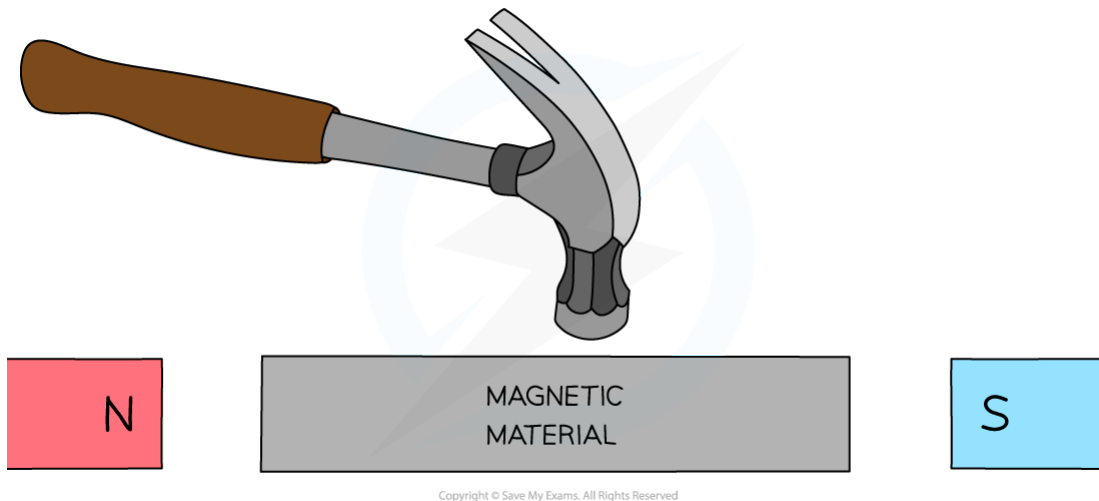
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Hitting with a hammer:

- If the material is placed in a magnetic field and then hit with a hammer, the material will also become magnetised



Hitting material placed into a magnetic field with a hammer can lead to the material becoming magnetised

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Methods of Demagnetisation

- Magnets can be demagnetised by using one of the following methods:
 - **Hit the material with a hammer** (when it is not in a magnetic field)
 - **Heat the material** (until it begins to glow) and then slowly let it cool
 - Place the material **in a coil containing alternating current (a.c.)** and then slowly withdraw the material (with the a.c. power source still attached to the coil)

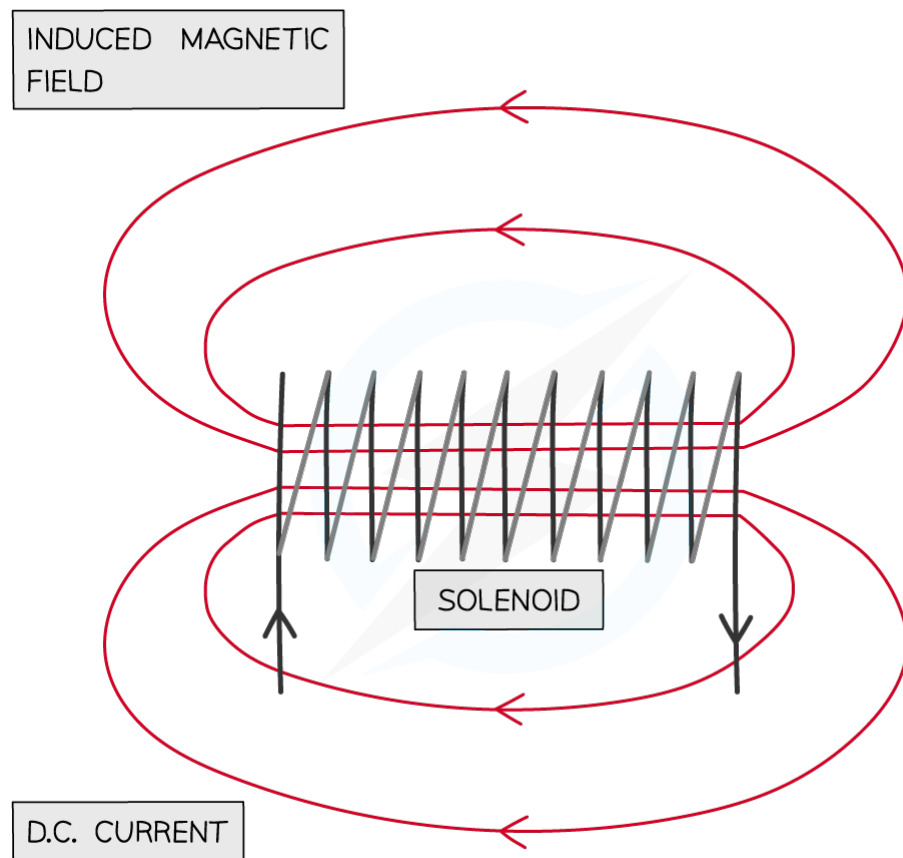
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Electromagnets

- When a current (d.c.) is passed through a long coil, the current creates a magnetic field around the coil similar to that of a bar magnet

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The magnetic field around a solenoid (a long coil) is identical to the magnetic field of a bar magnet

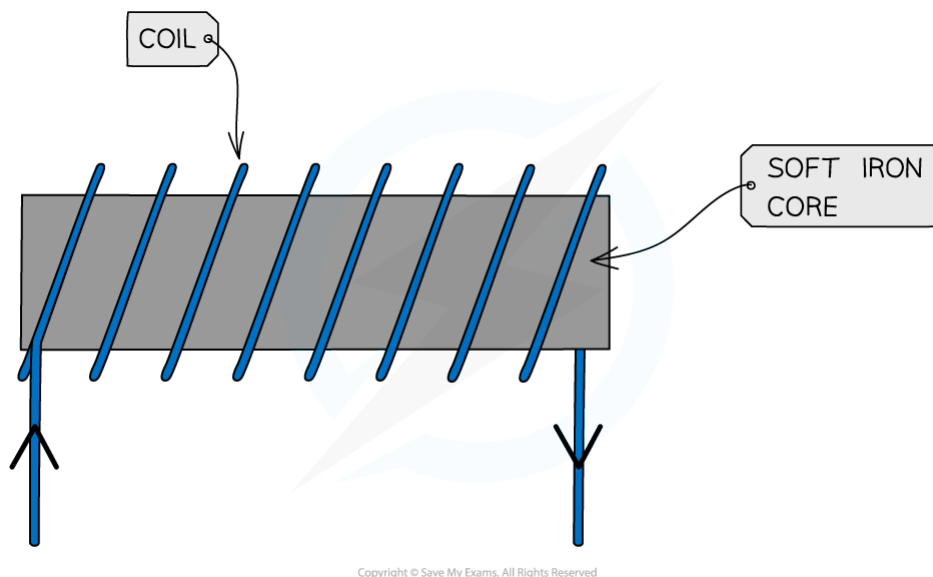
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- Placing a soft magnetic material (such as iron) inside the coil will make the field much stronger:

The coil becomes an electromagnet



An electromagnet consists of a coil of wire surrounding a soft iron core.

- Electromagnets have an advantage over permanent magnets as they can be turned on and off and the magnetic field strength can be changed
- The strength of an electromagnet's magnetic field may be increased by:
 - Increasing the current in the coil
 - Adding more turns to the coil

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Exam Tip

When discussing the strength of an electromagnet, avoid saying “add more coils”:

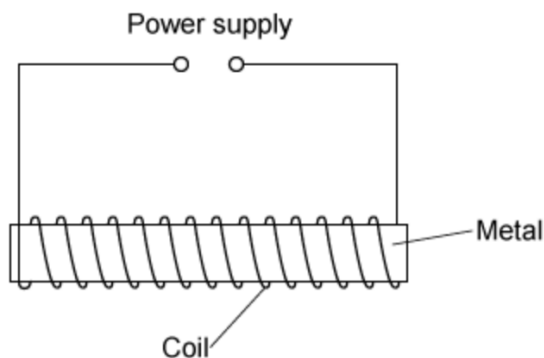
The coil describes the overall object – the individual loops of wire should be referred to as **turns**.

The correct phrase to use is “**add more turns to the coil**”.



Exam Question: Easy

A student wants to make an electromagnet using some easy-to-find materials that are in the school lab.



Which metal should the student use, and which setting on the power supply?

	metal	power supply setting
A	Soft iron	6V a.c.
B	steel	6V a.c.
C	Soft iron	6V d.c.
D	steel	6V d.c.

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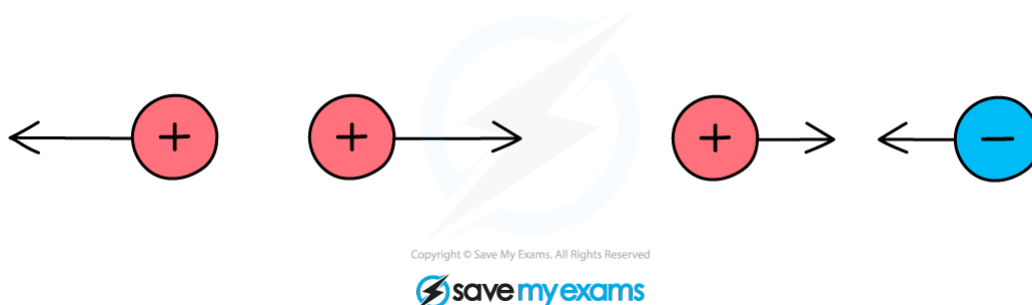


4.2 ELECTRICAL QUANTITIES

4.2.1 ELECTRIC CHARGE

Positive & Negative Charges

- Objects can be given one of two types of electric charge:
 - **Positive**
 - **Negative**
- When two charged objects are brought close together, there will be a force between those objects



Like charges repel; opposite charges attract

- Remember:
 - **Opposite charges attract**
 - **Like charges repel**



Exam Tip

Whilst electrostatic forces share many similarities with magnetic forces, they are different phenomena - **take care not to confuse the two!**

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Measuring Charge

- Electric charge is measured in units called **coulombs (C)**

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Demonstrating Electric Charge

- In your exam you will be expected to be able to explain how to give insulating materials a charge by the process of friction and to describe how to demonstrate the charging of materials in this way

Method:

- Suspend one of the insulating materials using a cradle and a length of string so that the material can rotate freely
- Rub one end of the material using a cloth (in order to give it a charge)
- Now take a second piece of insulating material and charge that by rubbing with a cloth
- Hold the charged end of the second piece close to the charged end of the first piece:
 - If the first piece rotates away (is repelled) from the second piece then the materials have the same charge
 - If the first piece moved towards (is attracted to) the second piece then they have opposite charges



Exam Tip

Experimental demonstrations, such as the one above, are different from experiments in which you have to take measurements.

When describing a demonstration you should **state a conclusion** – in other words, explain what you expect to happen and what it means.

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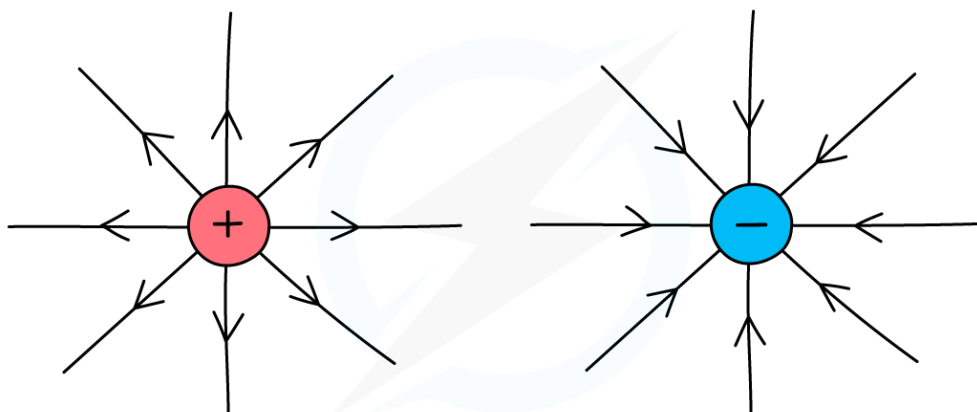


4.2.2 ELECTRIC FIELDS

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Simple Field Patterns

- Electric charges create electric fields in the regions surrounding them (Similar to the way in which magnets create magnetic fields)
- The electric field is the region in which another charge will experience a force
- Fields lines always go away from positive charges and towards negative charges – they have the same direction as the direction of the force on a positively charged particle at a point in that field

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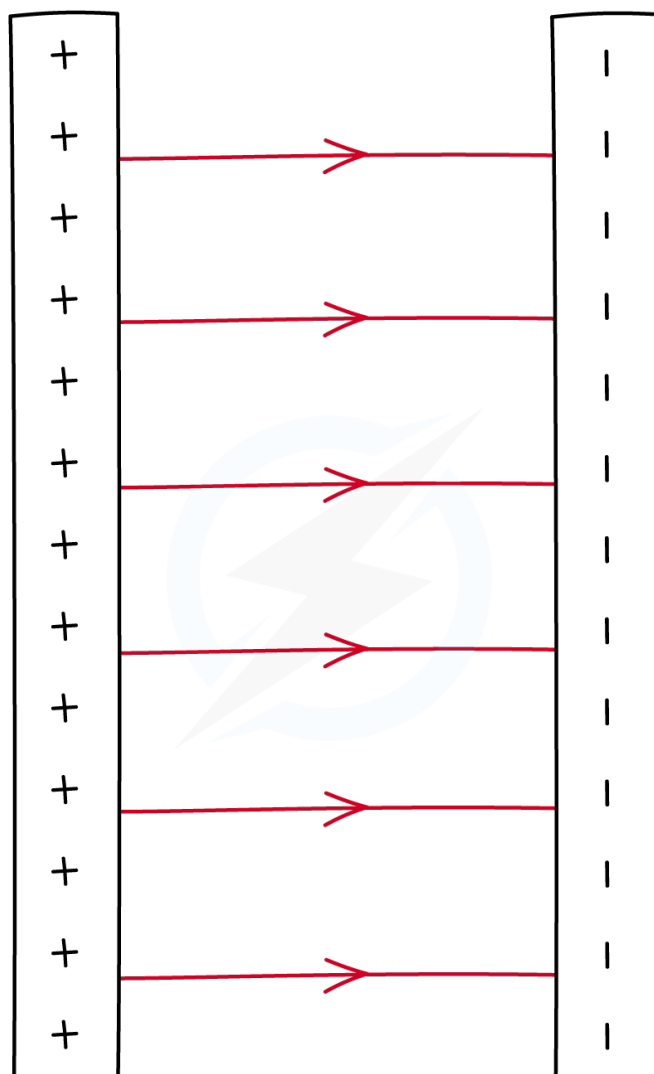
Electric fields are always directed away from positive charges and towards negative charges

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- Some simple field patterns that you ought to know:



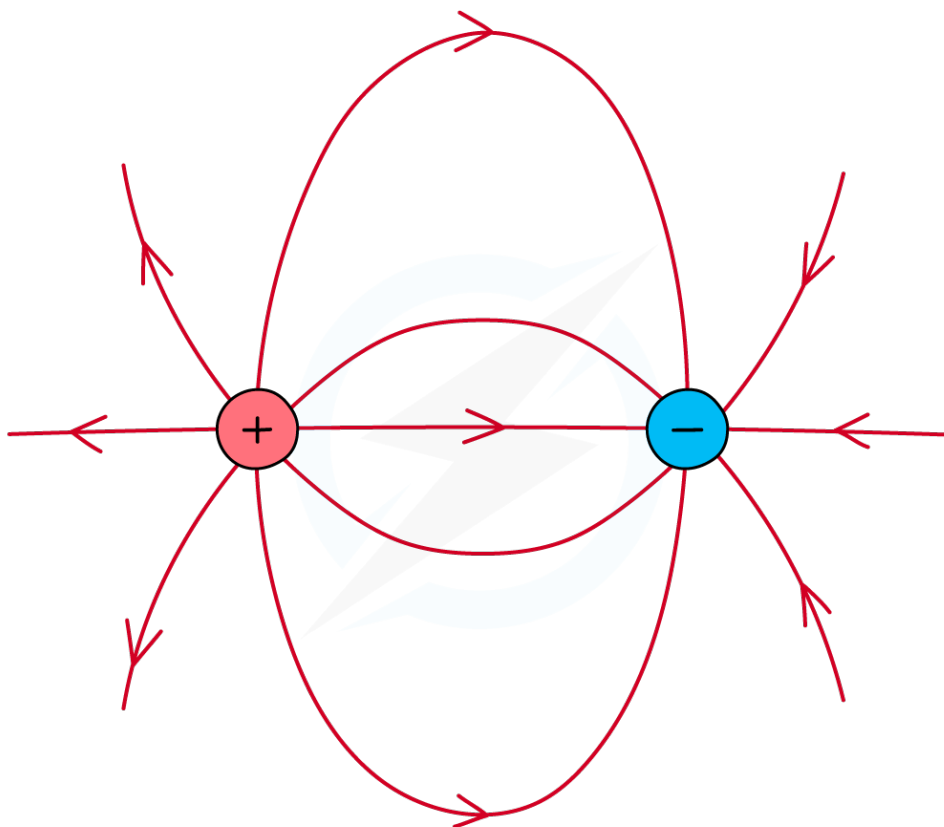
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The electric field between two parallel plates

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The electric field between two opposite charges

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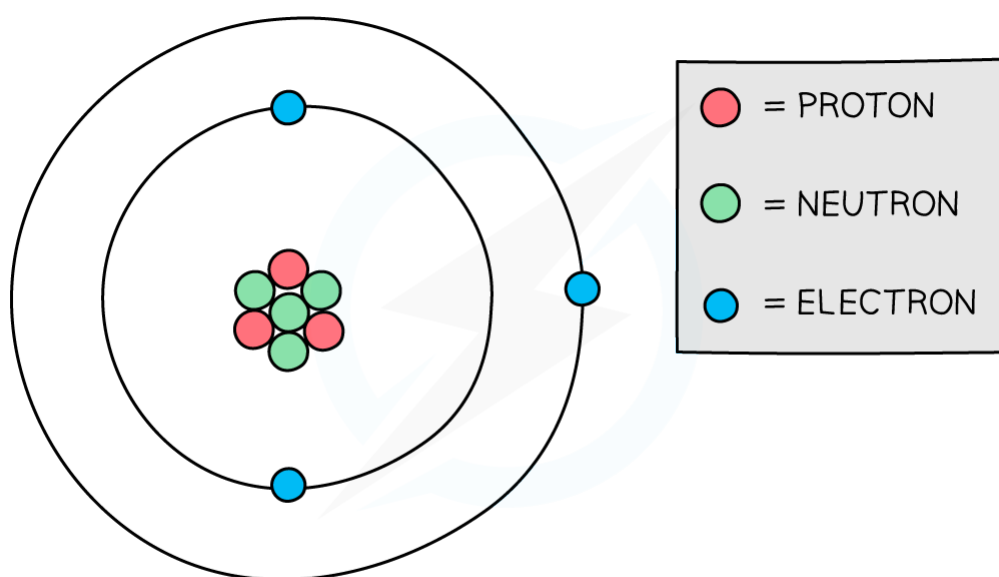
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4.2.3 ELECTRONS

Charge & Electrons

- **Electrons are negatively charged particles**
- Usually the number of electrons around an atom is equal to the number of (positive) protons in the nucleus of the atom, and hence the atom is neutral



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The number of negative electrons in an atom balances the number of positive protons

- If, however, an object gains extra electrons, it will gain an overall negative charge
- If it loses some of its electrons it will be left with a positive charge
 - **Negative charges are the result of gaining electrons**
 - **Positive charges are the result of losing electrons**

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Electrons & Conduction

- A conductor is a material that allows charge to flow through it
- An insulator is a material that does not allow the flow of charge through it

The table below lists some common conductors and insulators:

CONDUCTORS	INSULATORS
METALS	PLASTICS
GRAPHITE	GLASS
	MOST OTHER NON-METALS

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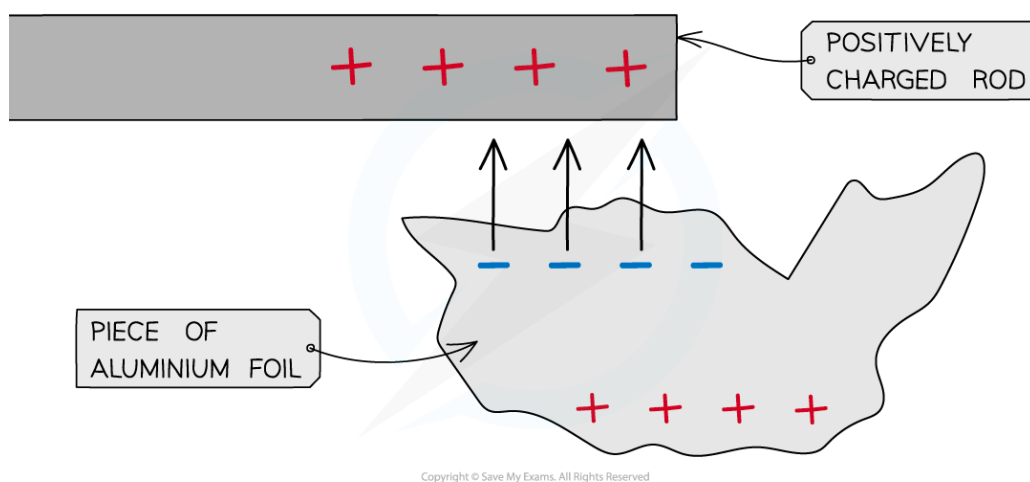


4.2.4 CHARGING BY INDUCTION

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Charging by Induction Explained

- When a charged object is held close to a conductor, electrons in the conductor are able to move towards (or away from) the charged object:



Negative electrons in the aluminium are attracted to the positive rod

- In the diagram above, electrons in the aluminium foil are attracted to the positively charged rod
- This causes the top of the foil to become negatively charged, whilst the bottom edge of it will be left with a positive charge
- The attraction between the positive rod and the negative charges on the top surface of the foil will cause the foil to be attracted to the rod

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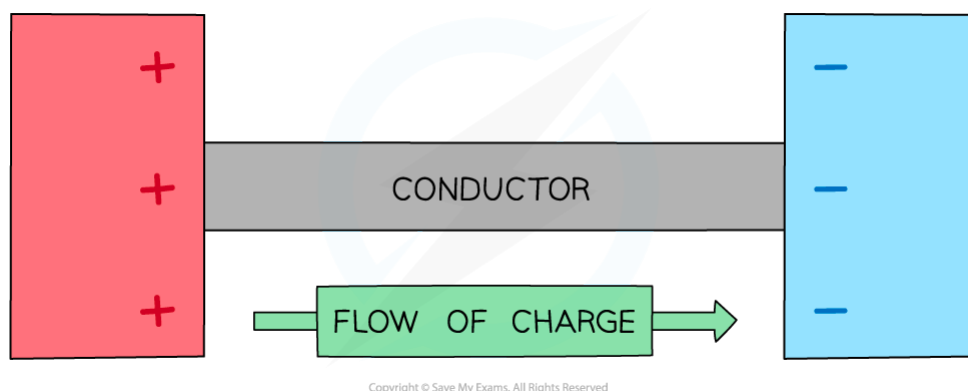
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4.2.5 CURRENT

Electric Current

- When two oppositely charged conductors are connected together (by a length of wire), charge will flow between the two conductors



Charge can flow between two conductors

- This flow of charge is called an **electric current**
 - The greater the flow of charge, the greater the electric current

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Charge, Current & Time

- **The current is the charge passing a point in a circuit every second**

(It is helpful to think of current as the **charge per second**)

- Charge, current and time are related by the following equation:

$$\text{CHARGE} = \text{CURRENT} \times \text{TIME}$$

$$Q = I \times t$$

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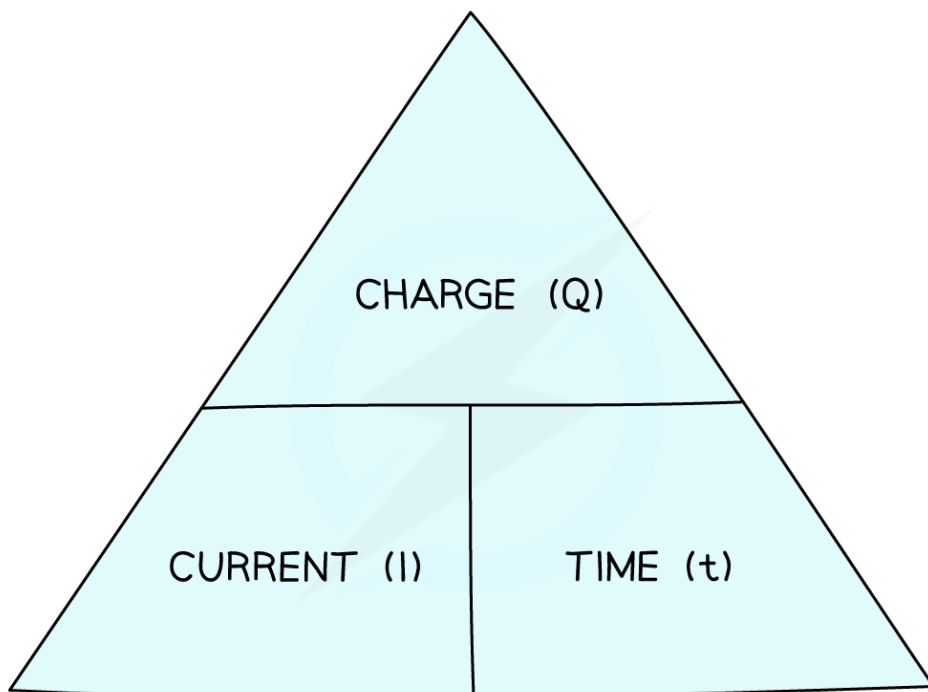
- Where the symbols:
 - **Q** stands for charge (measured in coulombs, C)
 - **I** stands for current (measured in amps, A)

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- You can rearrange this equation with the help of the formula triangle:



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Use the formula triangle to help you rearrange the equation

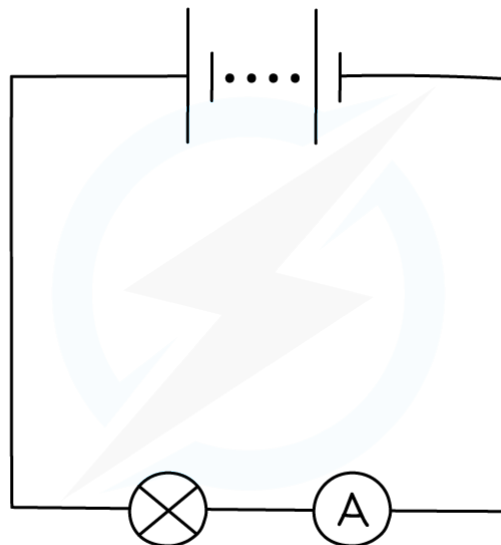
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Measuring Current

- Current is measured using an ammeter
- Ammeters should always be connected in series with the part of the circuit you wish to measure the current through



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An ammeter can be used to measure the current around a circuit

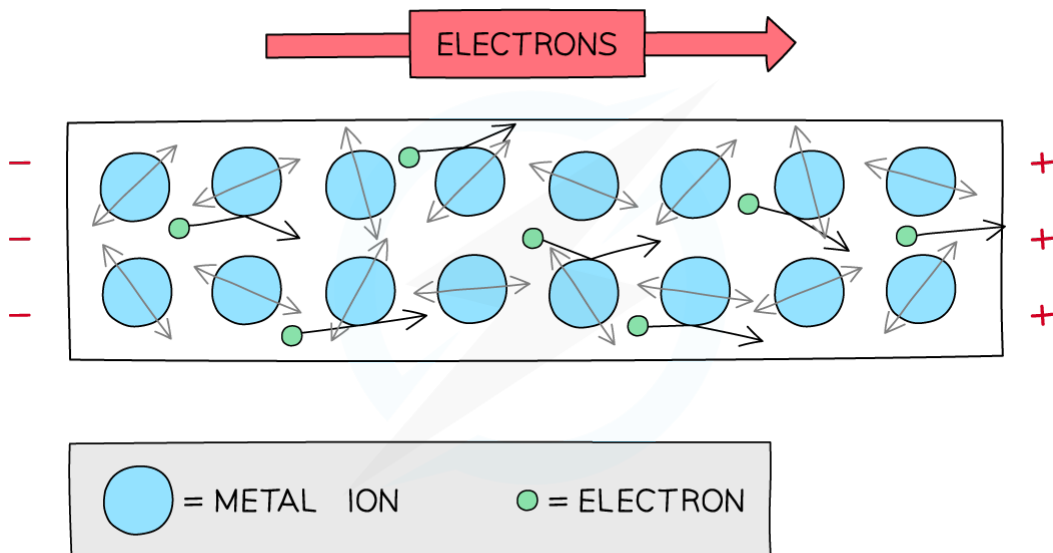
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Current & Electrons

- In a metal, current is caused by a flow of electrons



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In metals, the current is caused by a flow of free (delocalised) electrons

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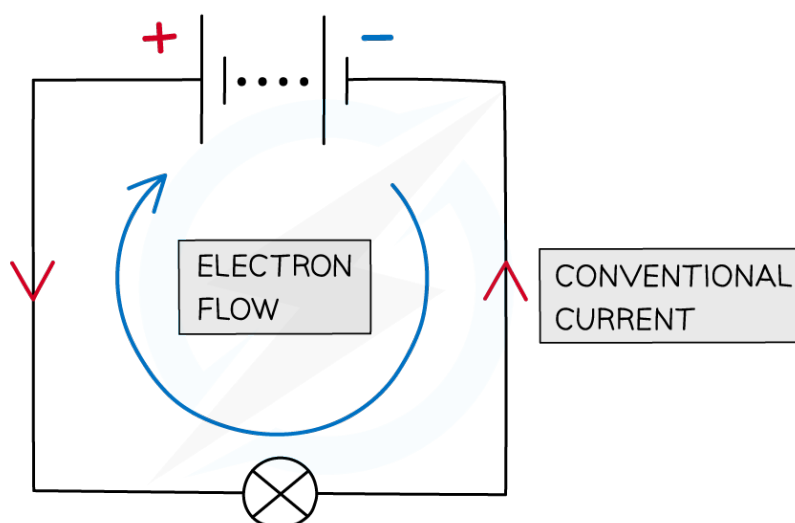
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Electrons & Conventional Current

- **Electrons are negatively charged**
- This means that the **electrons flow from negative to positive**
- **Conventional current**, however, is still defined as going from **positive to negative**

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By definition, conventional current always goes from positive to negative (even though electrons go the other way)

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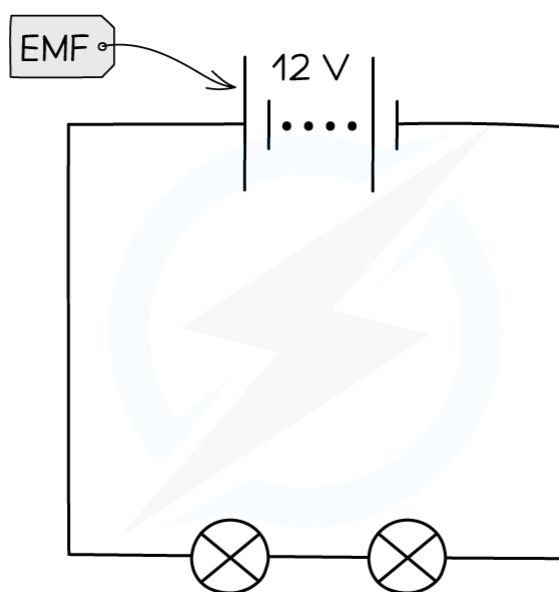
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4.2.6 ELECTROMOTIVE FORCE

What is Electromotive Force?

- The Electromotive Force (EMF) is the name given to the Potential Difference (Voltage) of the power source in a circuit
- The Electromotive Force (EMF) is measured in Volts (V)



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The EMF is the voltage supplied by a power supply: 12 V in the above case

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EMF & Energy

- The EMF of a power supply (measured in volts, V) is the amount of energy (measured in joules, J) supplied to each coulomb of charge passing through that power supply



Exam Tip

Think of the EMF as being the **energy per coulomb**.

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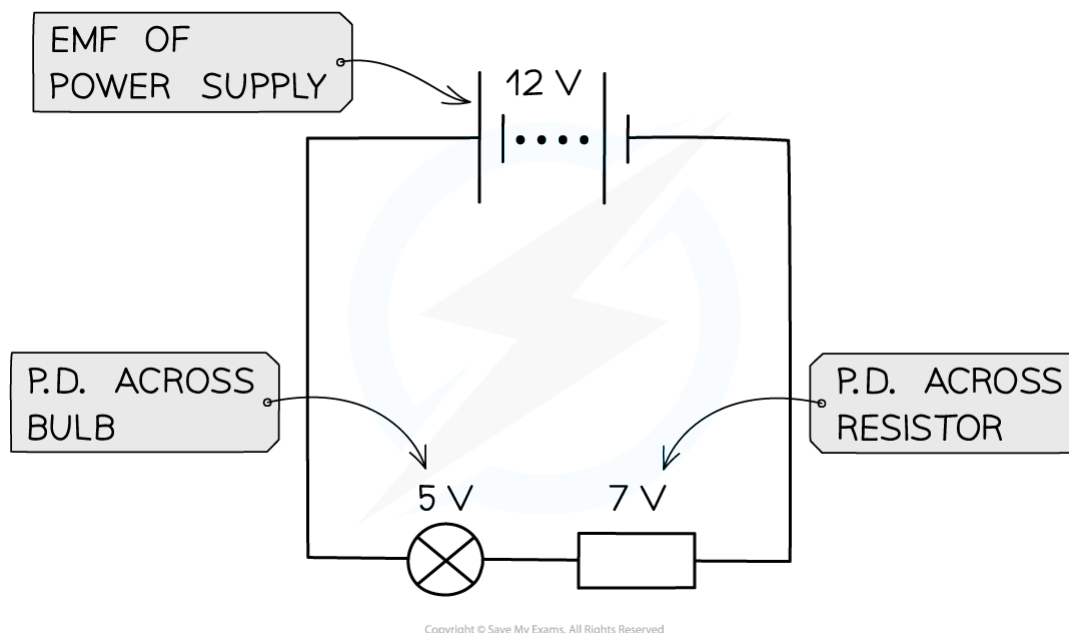
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4.2.7 POTENTIAL DIFFERENCE

What is Potential Difference?

- As charge flows around a circuit, energy is transferred to or from the charge
- The potential difference (voltage) between two points in a circuit is related to the amount of energy transferred between those points in the circuit
- Potential difference is measured in volts (V)



The potential difference is the voltage across each component: 5 volts for the bulb (on the left) and 7 volts for the resistor (on the right)

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Potential Difference & Energy

- **The potential difference between two points in a circuit is the amount of energy transferred by each unit of charge passing between those two points**
- The unit of voltage, **the volt (V)**, is the same as a joule per coulomb (J/C)

$$1 \text{ V} = 1 \text{ J/C}$$

- So, for example:
 - If a bulb has a voltage of 3 V, every coulomb of charge passing through the bulb will lose 3 J of energy



Exam Tip

Think of potential difference as being the **energy per coulomb**.

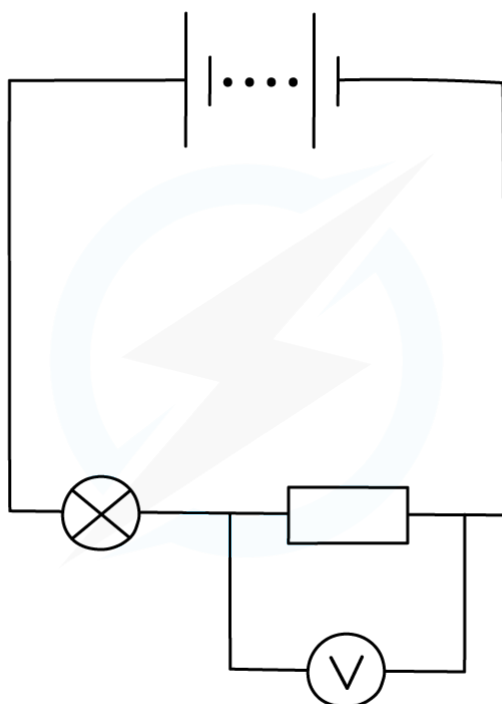
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Measuring Potential Difference

- Potential difference can be measured using a **voltmeter**
- The voltmeter should be **connected in parallel** with the part of the circuits you want to measure the potential difference of



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Potential difference can be measured by connecting a voltmeter between two points in a circuit

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4.2.8 RESISTANCE

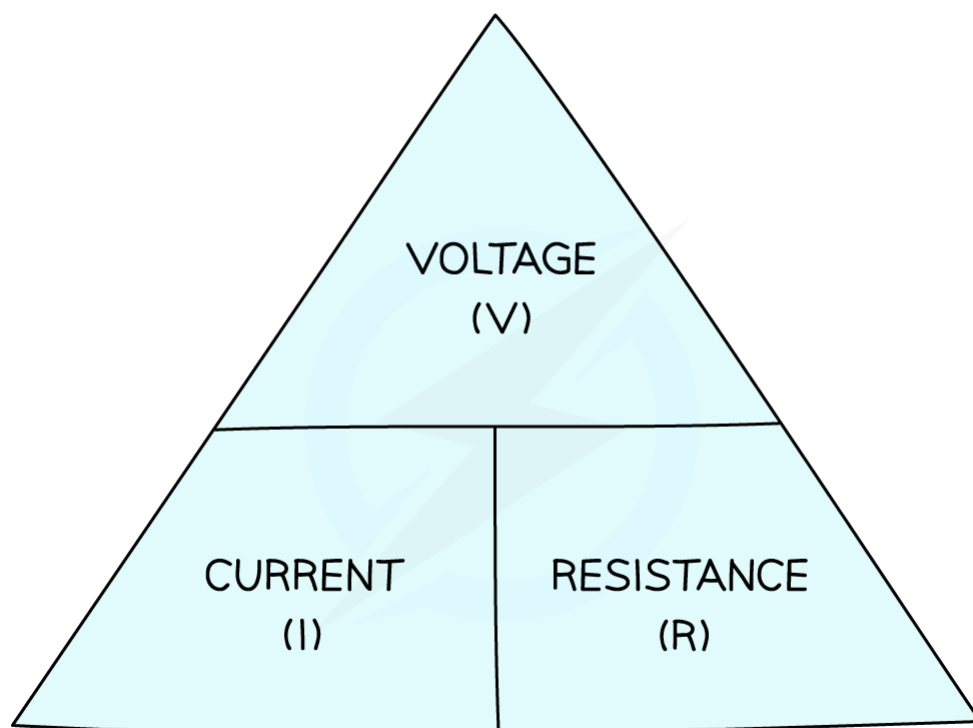
What is Resistance?

- **Resistance is the opposition to current**
 - For a given potential difference:

The higher the resistance, the lower the current

- Potential difference, current and resistance are related by the following equation:

$$\text{POTENTIAL DIFFERENCE} = \text{CURRENT} \times \text{RESISTANCE}$$

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Use the formula triangle to help you rearrange the equation

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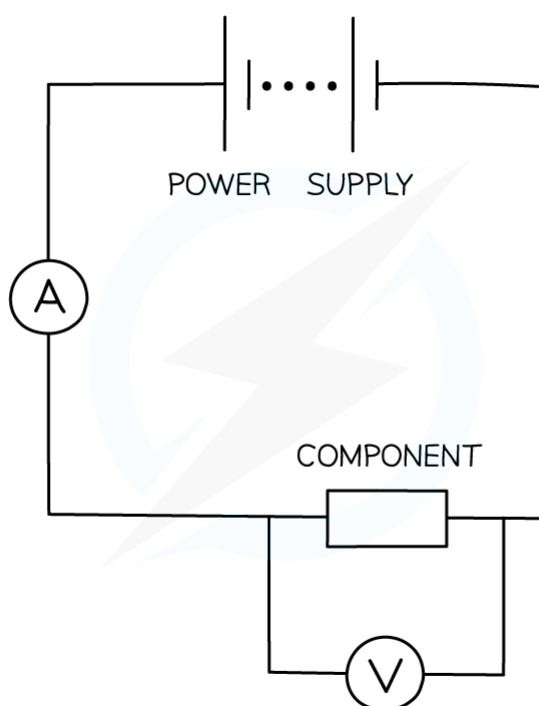
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- The unit of resistance is the ohm (Ω)

Determining Resistance

- To find the resistance of a component, set up a circuit like the one shown below



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A circuit to determine the resistance of a component

- The power supply should be set to a low EMF (voltage) to avoid heating the component – 1 or 2 volts is typically enough
- Measurements of the potential difference and current should then be taken from the voltmeter and ammeter respectively
- Finally, these readings should be substituted into the following equation:

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$$\text{resistance} = \frac{\text{potential difference}}{\text{current}}$$

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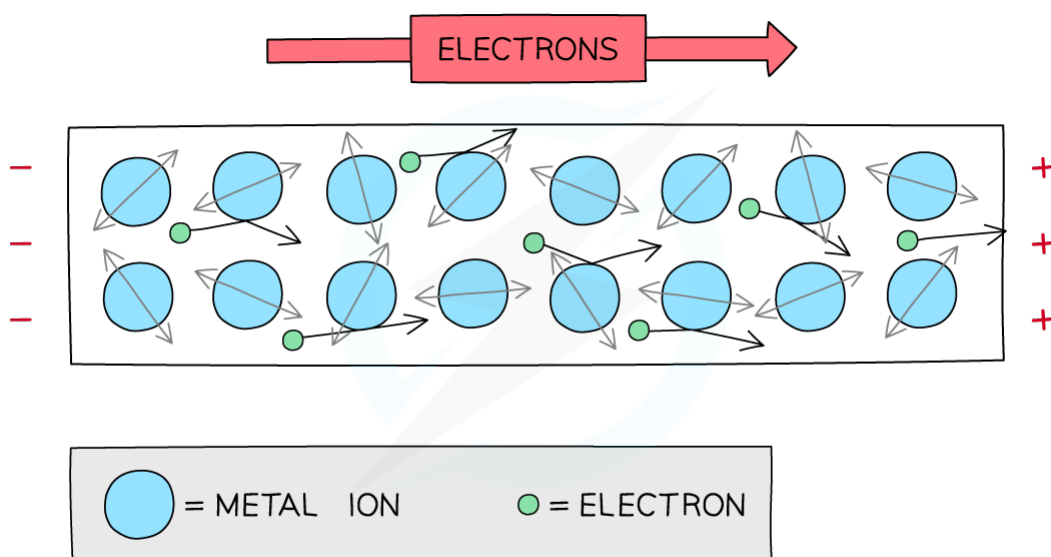
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4.2.9 THE RESISTANCE OF A WIRE

Resistance of a Wire: Basics

- As electrons pass through a wire, they collide with the metal ions in the wire



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Electrons collide with ions, which resist their flow

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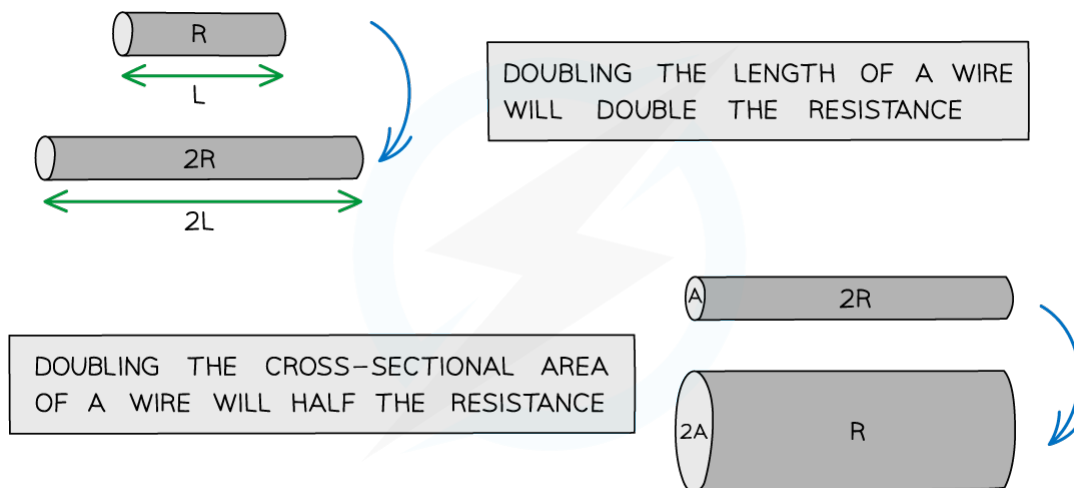


- The ions get in the way of the electrons, resisting their flow
- If the wire is longer, each electron will collide with more ions and so there will be more resistance:

The longer a wire, the *greater* its resistance

- If the wire is thicker (greater diameter) there is more space for the electrons and so more electrons can flow:

The thicker a wire, the *smaller* its resistance



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The length and width of the wire affect the resistance

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Resistance of a Wire

- **The resistance of a wire is proportional to its length**
- This means that if the length of a wire is doubled, its resistance will double
- **The resistance of a wire is inversely proportional to its cross-sectional area**
- This means that if the cross-sectional area of a wire is doubled, its resistance will halve



Exam Tip

Cross-sectional area is proportional to the diameter squared.

This means that if the diameter is doubled, the cross-sectional area will quadruple, causing the resistance to drop to a quarter.

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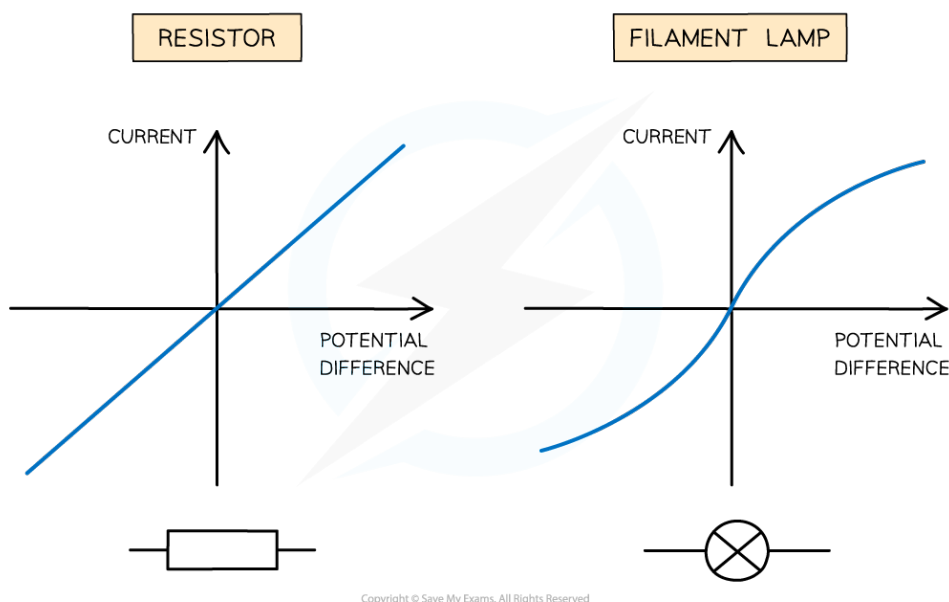


4.2.10 IV GRAPHS

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Current & Potential Difference

- As the potential difference (voltage) across a component is increased, the current in the component also increases
- The precise relationship between voltage and current can be different for different types of components and is shown by an IV graph:



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IV graphs for a resistor and a filament lamp

- The IV graph **for a resistor** is very simple:
The current is proportional to the potential difference
- This is because the resistor has a constant resistance

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- **For a lamp** the relationship is more complicated:
The current increases at a proportionally slower rate than the potential difference
- This is because:
 - The current causes the filament in the lamp to heat up
 - As the filament gets hot its resistance increases
 - This opposes the current, causing it to increase at a slower rate

4. Electricity & Magnetism

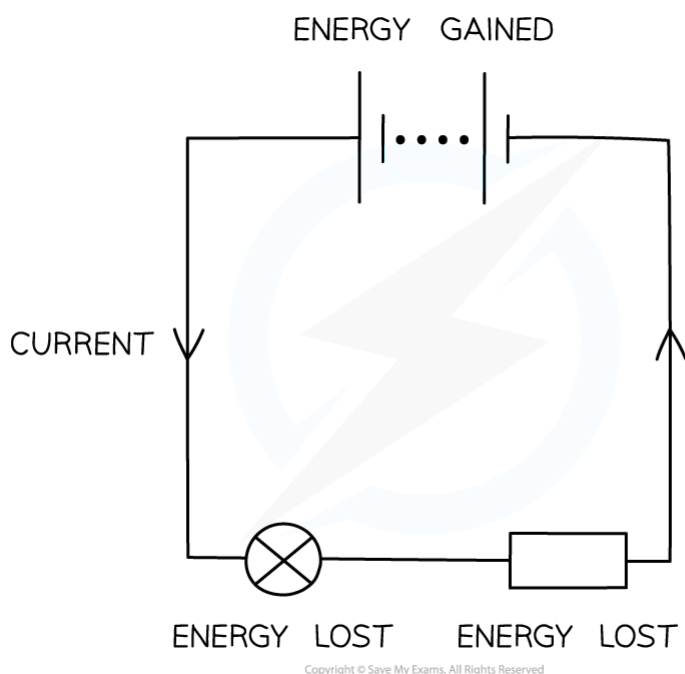
YOUR NOTES



4.2.11 ELECTRICAL WORKING

Electrical Energy

- As electricity passes around a circuit, energy is transferred from the power source to the various components (which may then transfer energy to the surroundings)
 - As charge passes through the power supply it is given energy
 - As it passes through each component it loses some energy (transferring that energy to the component)



The current transfers electrical energy from the power source and to the components

4. Electricity & Magnetism

YOUR NOTES



Extended Only

Calculating Electrical Energy

- The amount of electrical energy used by a component or appliance depends upon three things:
 - The current
 - The potential difference
 - The amount of time the device is used for
- The energy transferred can be calculated from the equation:

energy transferred = current × potential difference × time

$$E = I \times V \times t$$

- Where **the unit of energy is the joule (J)**

4. Electricity & Magnetism

YOUR NOTES



Extended Only

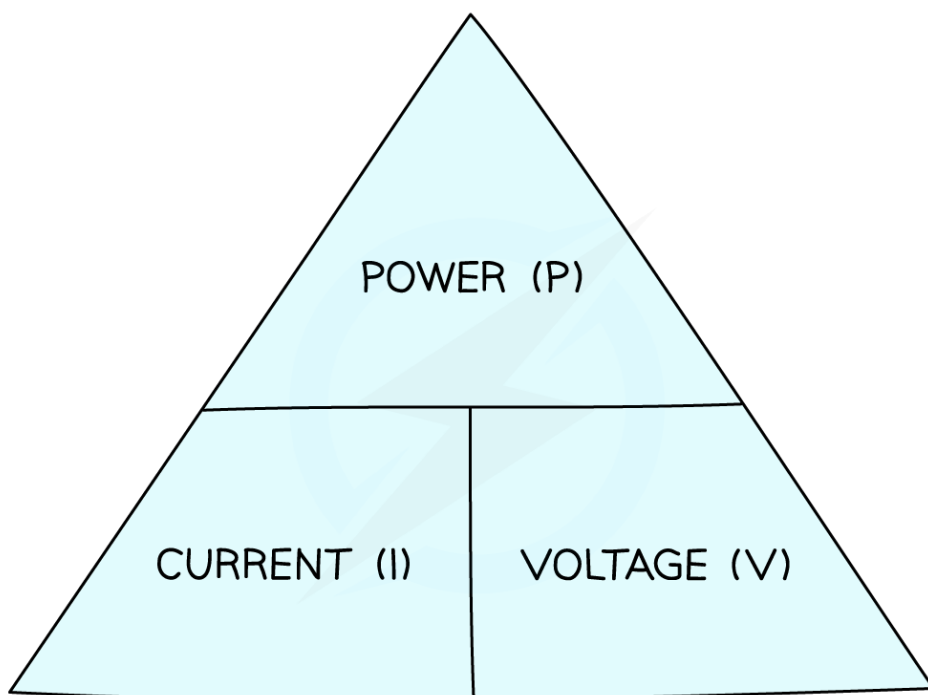
Electrical Power

- **Power is the rate of energy transfer** (the amount of energy transferred per second)
- The power of an electrical component (or appliance) is given by the equation:

Power = Current $\times$ potential difference

$$P = I \times V$$

- You can rearrange this equation with the help of the formula triangle:



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Use the formula triangle to help you rearrange the equation

- The unit of power is the **watt (W)**, which is the same as a **joule per second (J/s)**
- Note that the above equation is related to the equation for energy – after all, **power is just energy per second**, and so:

$$\begin{array}{l} \text{energy transferred} \\ \text{difference} \times \text{time} \end{array} = \text{power} \times \text{time} = \text{current} \times \text{potential difference} \times \text{time}$$

4. Electricity & Magnetism

YOUR NOTES



Exam Tip

Remember: Power is just **energy per second**.

"Watt is the unit of power?"



Exam Question: Easy

What is the definition of an electric field?

- A** A region in space in which a mass experiences a force due to the Earth's mass.
- B** A region in space through which electromagnetic radiation is passing.
- C** A region in space in which a compass needle experiences a force.
- D** A region in space in which an electric charge experiences a force.

4. Electricity & Magnetism

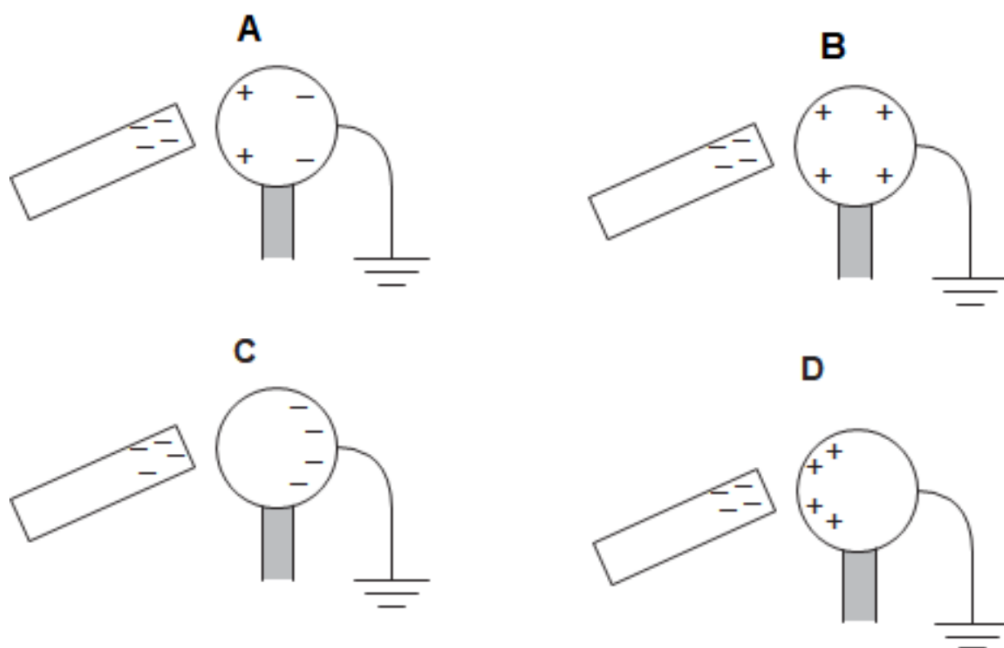
YOUR NOTES



? Exam Question: Hard

A student rubs a polythene rod with a cloth, giving it a negative charge. She then holds it near an earthed, conducting sphere as shown in the diagrams below.

Which of the diagrams shows the correct distribution of charges on the conducting sphere?



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4. Electricity & Magnetism

YOUR NOTES



4.3 ELECTRIC CIRCUITS

4.3.1 CIRCUIT DIAGRAMS

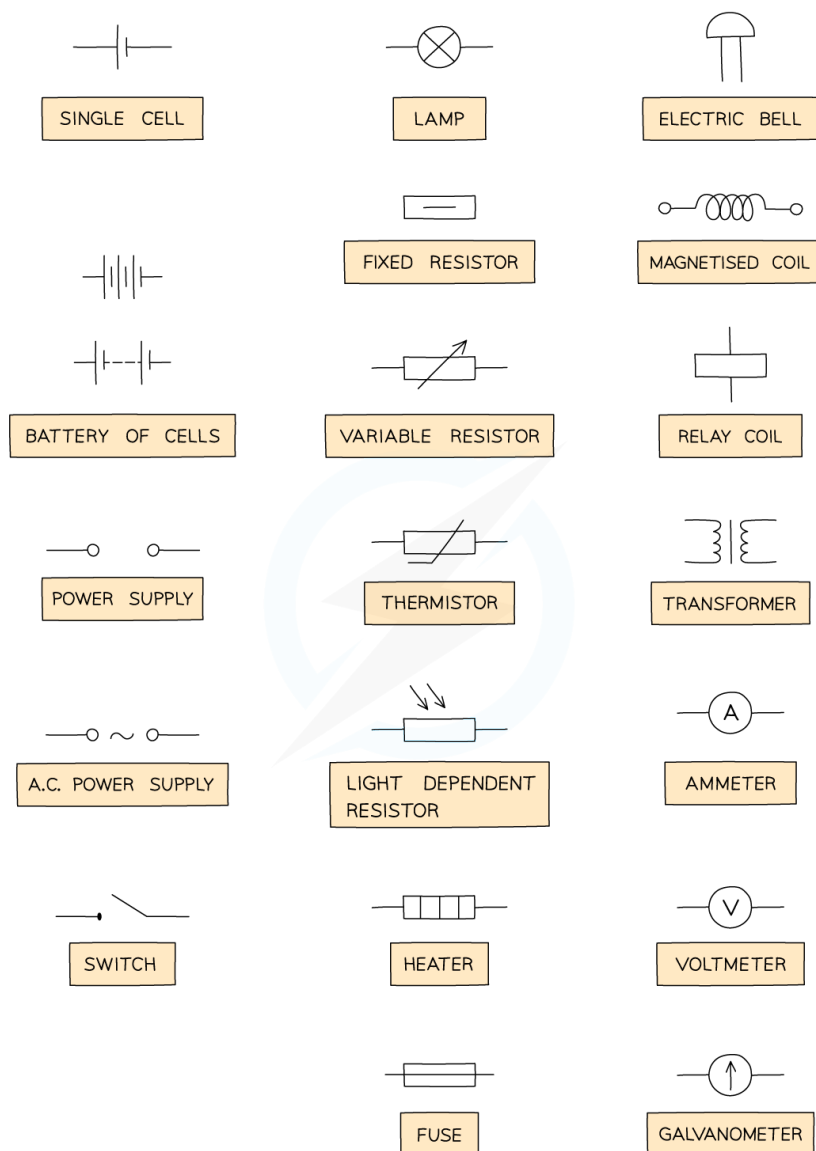
4. Electricity & Magnetism

YOUR NOTES



Standard Circuit Symbols

- The diagram below shows the various circuit symbols that could be used in circuit diagrams. You will be expected to know what each one is



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You are expected to be able to recognise and draw the above symbols

4. Electricity & Magnetism

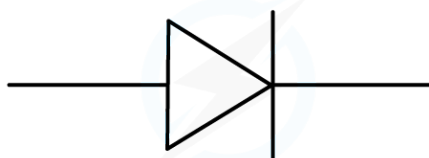
YOUR NOTES



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Diode Circuit Symbol

- In addition to the above, you should be able to recognise and draw the circuit symbol for a diode:

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A diode is a component that only allows a current in one direction

(Note: diodes are occasionally drawn with a horizontal line running through the middle of them)

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YOUR NOTES



4.3.2 SERIES & PARALLEL CIRCUITS

Series Circuits

- A series circuit consists of a string of two or more components, connected end to end:

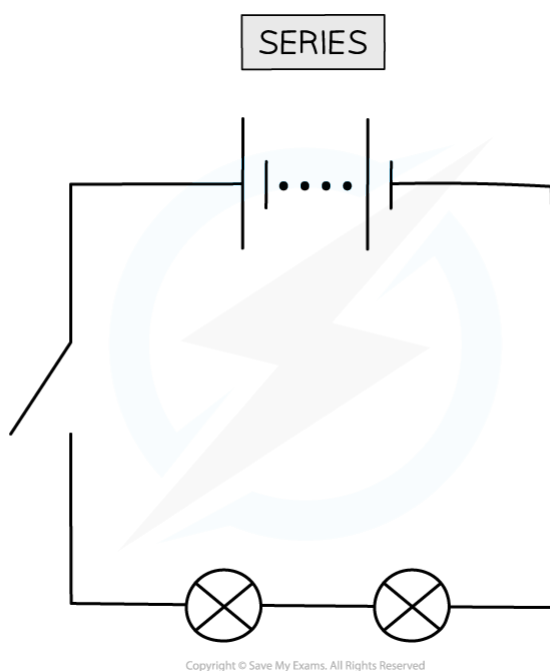


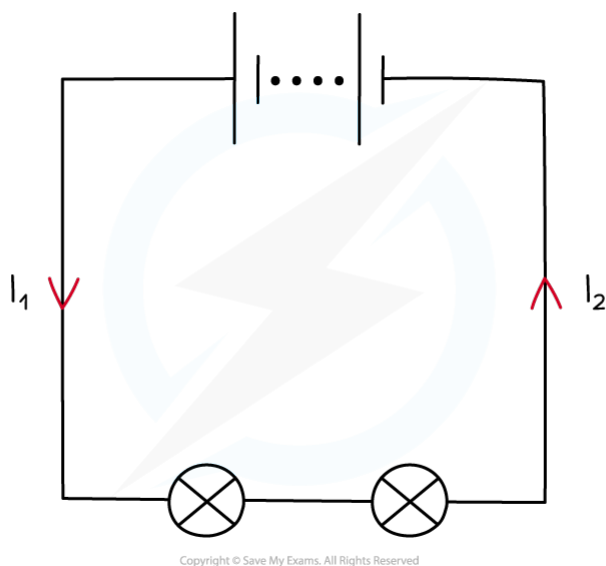
Diagram showing two bulbs connected in series

4. Electricity & Magnetism

YOUR NOTES



- In a series circuit the current is the same at all points



The current is the same at all points in a series circuit

4. Electricity & Magnetism

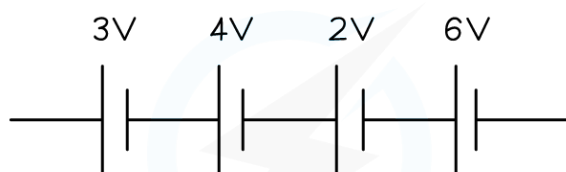
YOUR NOTES



Extended Only

Potential Difference in Series

- When several cells are connected together in series, their combined EMF is equal to the sum of their individual EMFs



$$\text{TOTAL EMF} = 3 + 4 + 2 + 6 = 15 \text{ V}$$

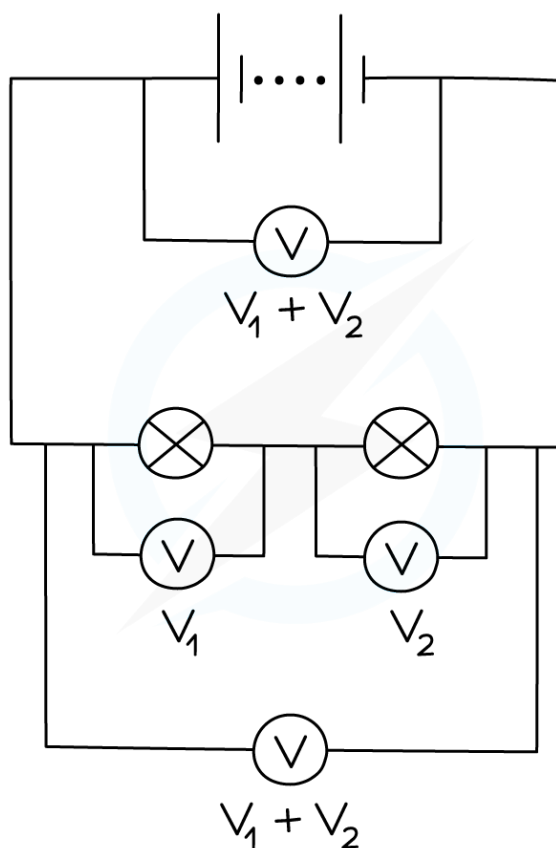
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The total EMF of these cells is equal to the sum of their individual EMFs

- In a series circuit, the sum of potential differences across the components is equal to the total EMF of the power supply

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YOUR NOTES



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In a series circuit the components share the EMF of the power supply

4. Electricity & Magnetism

YOUR NOTES



Parallel Circuits

- A parallel circuit consists of two or more components attached along separate branches of the circuit

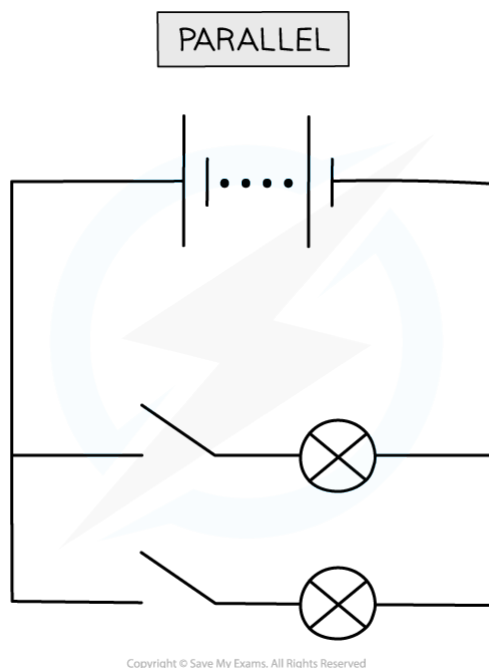


Diagram showing two bulbs connected in parallel

- The **advantages** of this kind of circuit are:
 - **The components can be individually controlled, using their own switches**
 - **If one component stops working the others will continue to function**
- **In a parallel circuit, the current splits up** – some of it going one way and the rest going the other
- This means that the current in each branch will be smaller than the current from the power supply

4. Electricity & Magnetism

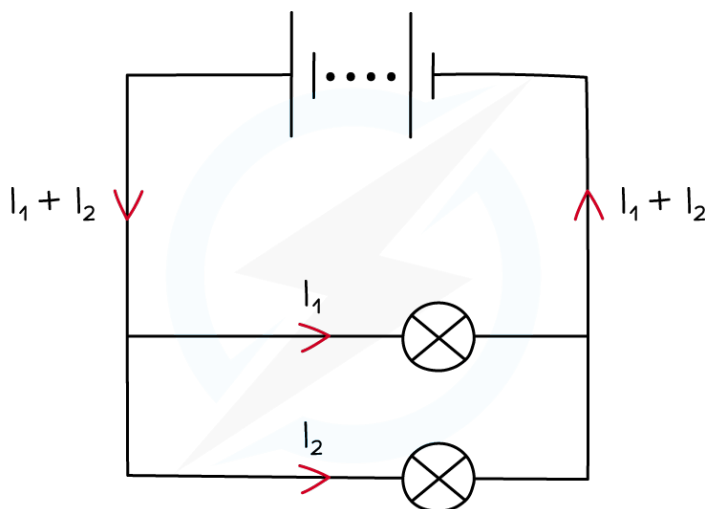
YOUR NOTES



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Determining Current in Parallel

- Because the current splits up, the sum of currents in each branch will equal the current from the power supply



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In a parallel circuit, the current splits up, dividing between the various branches of the circuit

- Note that the current does not always split equally - often there will be more current in some branches than in others
- The current in each branch will only be identical if the components along each branch are identical (or at least have the same resistance)

4. Electricity & Magnetism

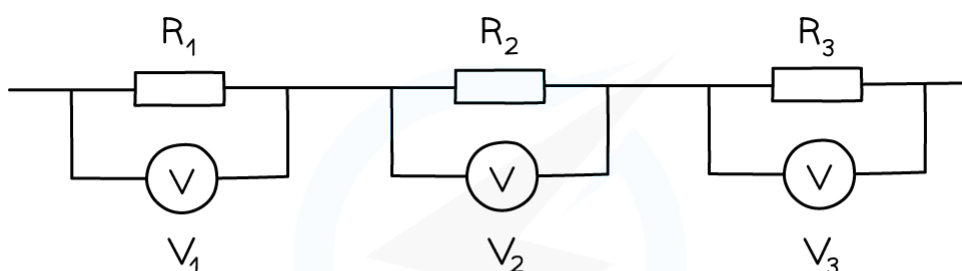
YOUR NOTES



4.3.3 RESISTORS IN SERIES & PARALLEL

Resistors in Series

- When two or more components are connected in series:
 - The combined resistance of the components is equal to the sum of individual resistances**



$$\text{TOTAL VOLTAGE} = V_1 + V_2 + V_3$$

$$\text{COMBINED RESISTANCE} = R_1 + R_2 + R_3$$

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When several components are connected in series, their combined resistance is equal to the sum of their individual resistances

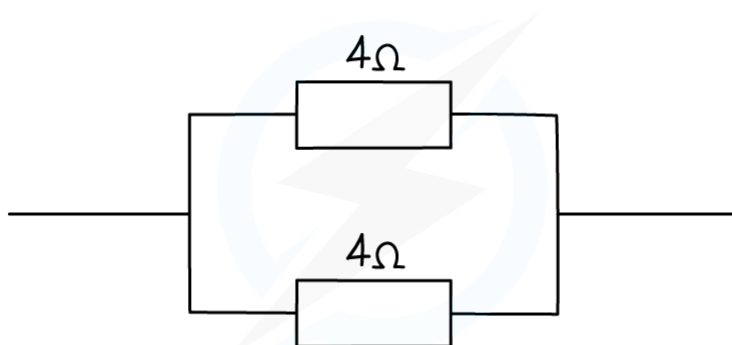
4. Electricity & Magnetism

YOUR NOTES



Resistors in Parallel

- When resistors are connected in parallel, the combined resistance decreases and is less than the resistance of any of the individual components
- If two resistors of equal resistance are connected in parallel, then the combined resistance will halve



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The above resistors will have a combined resistance of $2\ \Omega$ – half the value of each resistor

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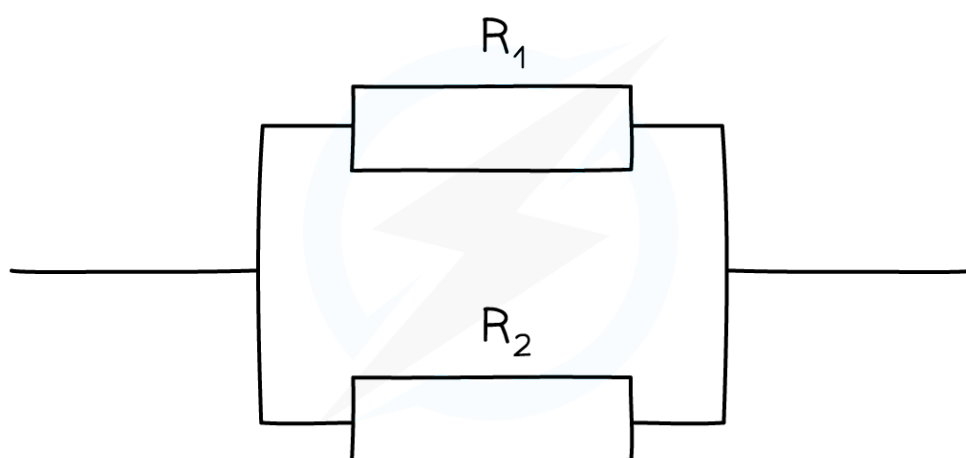
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Determining Resistance in Parallel

- More generally, to determine the combined resistance of any combination of two resistors, you must use the equation:



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$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

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The above equation is not the same as $R = R_1 + R_2$ - a common, incorrect simplification that is made

- To calculate the resistance:
 - First find the value of $1/R$ (by adding $1/R_1 + 1/R_2$)
 - Next find the value of R by using the reciprocal button on your calculator (labelled either x^{-1} or $1/x$, depending on your calculator)

4. Electricity & Magnetism

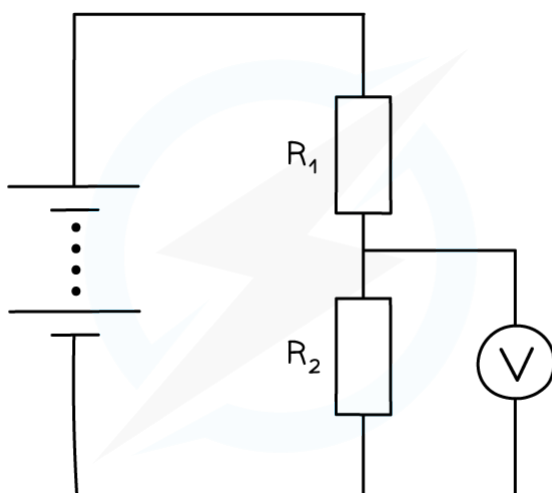
YOUR NOTES



4.3.4 USE OF CIRCUIT COMPONENTS

Potentiometers

- When two resistors are connected in series, the potential difference across the power source is shared between them

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A potential divider splits the potential difference of a power source between two components

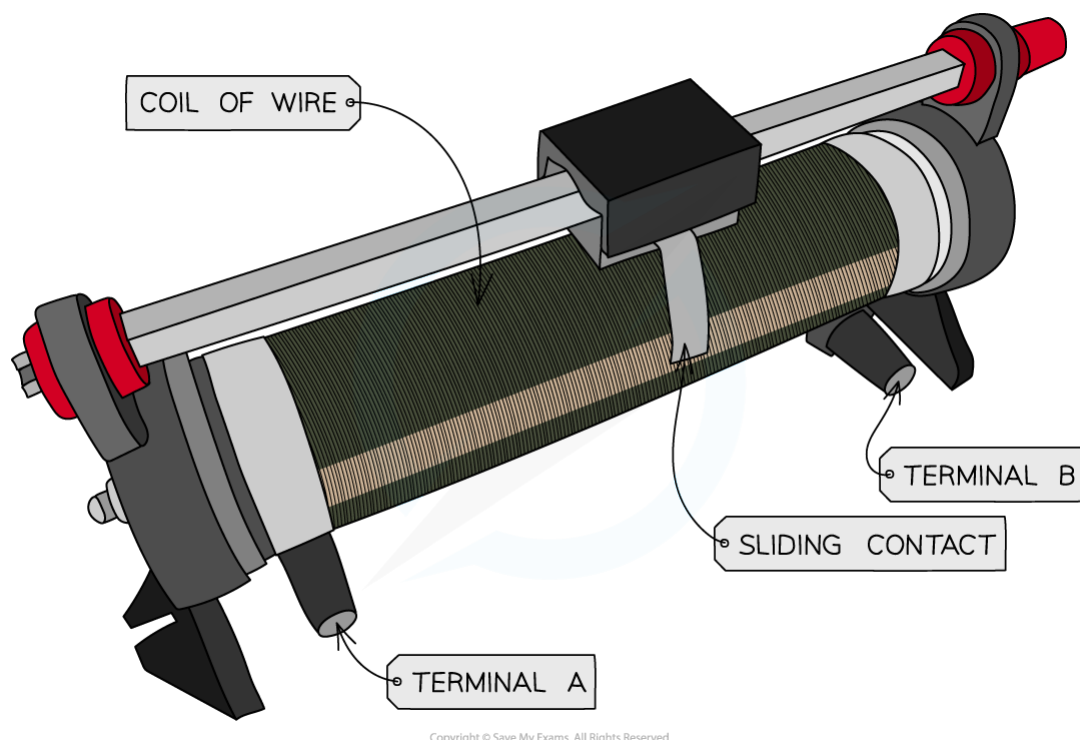
- The potential difference across each resistor depends upon its resistance:
 - The resistor with the largest resistance will have a greater potential difference than the other one
 - If the resistance of one of the resistors is increased, it will get a greater share of the potential difference, whilst the other resistor will get a smaller share

4. Electricity & Magnetism

YOUR NOTES



- A potentiometer is a single component that (in its simplest form) consists of a coil of wire with a sliding contact, midway along it



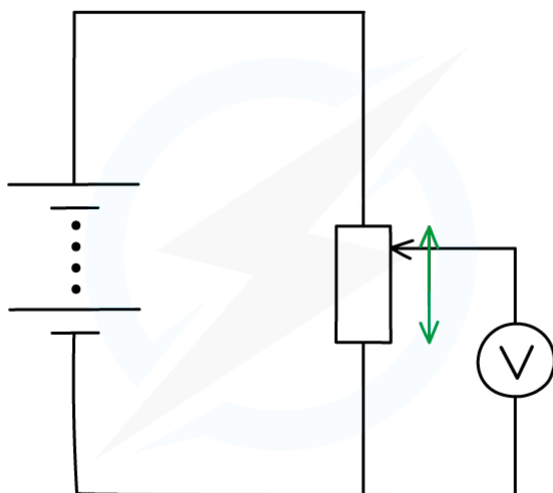
A potentiometer is a kind of variable resistor

4. Electricity & Magnetism

YOUR NOTES



- The sliding contact has the effect of separating the potentiometer into two parts – an upper part and a lower part – both of which have different resistances



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Moving the slider (the arrow in the diagram) changes the resistances (and hence potential differences) of the upper and lower parts of the potentiometer

- If the slider in the above diagram is moved upwards, the resistance of the lower part will increase and so the potential difference across it will also increase

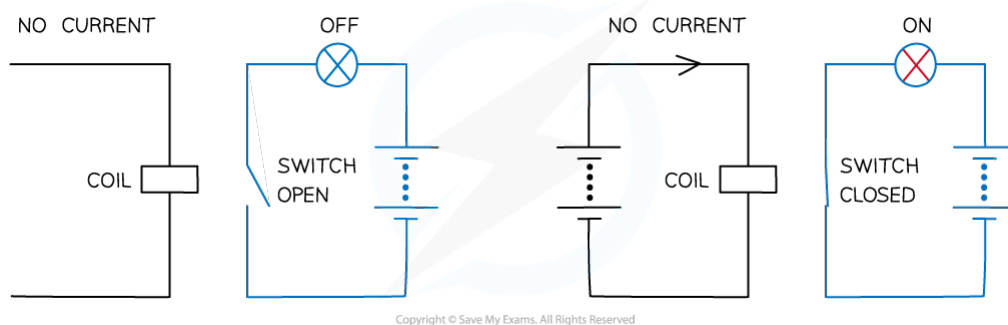
4. Electricity & Magnetism

YOUR NOTES



Relay

- A relay consists of two parts:
 - A coil (electromagnet)
 - A magnetic switch



When there is a current in the coil, it attracts the switch which completed the right-hand part of the circuit

- When there is a current in the coil, it creates a magnetic field which attracts the switch, closing it
- This can be used to control a switch in a separate circuit, as shown in the above diagram

Thermistors & LDRs



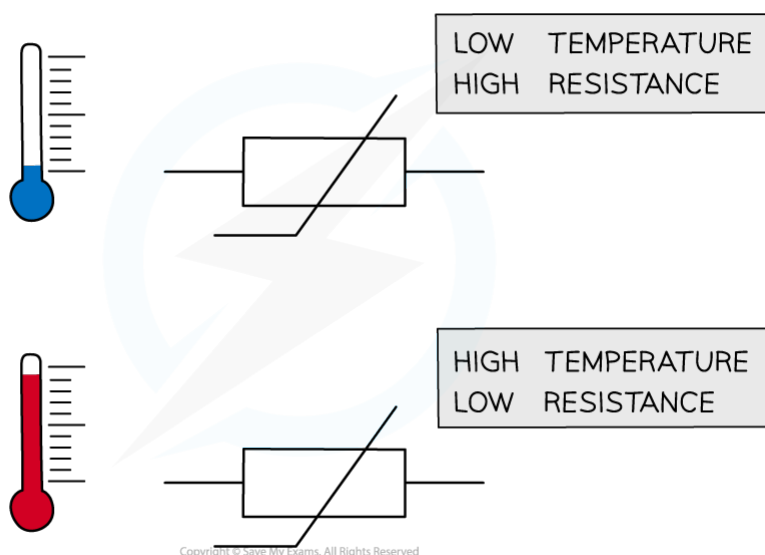
4. Electricity & Magnetism

YOUR NOTES



The symbol for a thermistor

- A thermistor is a type of resistor (a thermal resistor) which changes its resistance when it is heated up:
 - As a thermistor gets hotter, its resistance decreases
 - As it gets colder, its resistance increases



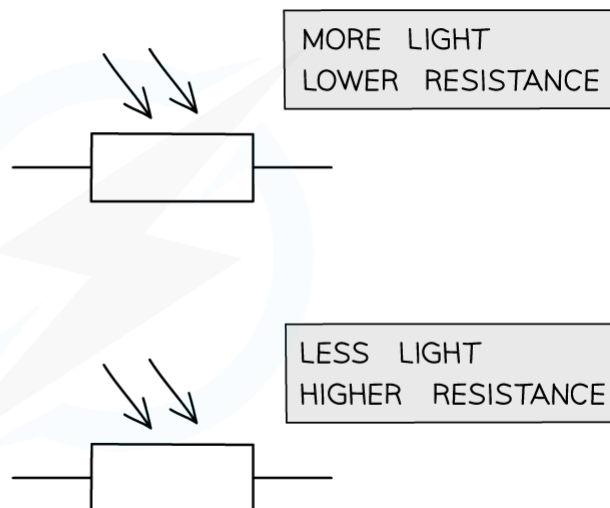
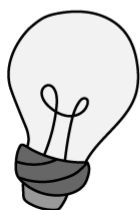
The resistance through a thermistor is dependent on the temperature of it

4. Electricity & Magnetism

YOUR NOTES



LDRs



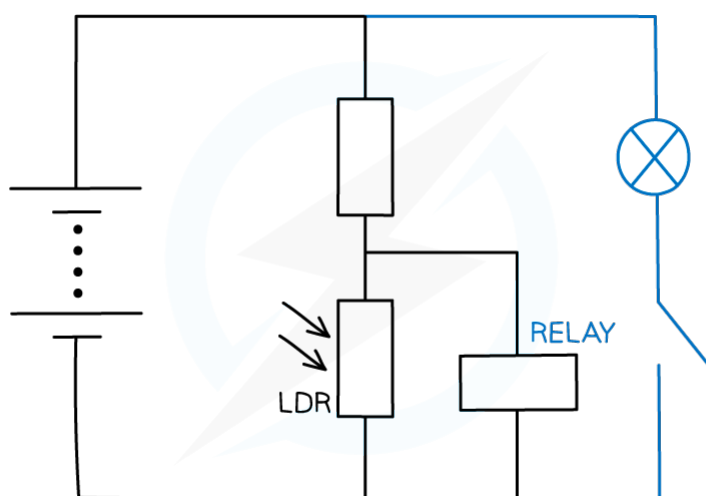
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The resistance through an LDR is dependent on the intensity of light falling on it

- An LDR (Light Dependent Resistor) is a type of resistor which changes its resistance when light is shone on it:
 - When more light is shone on an LDR its resistance decreases
 - If an LDR is covered up (reducing the light hitting it) its resistance increases.
- Note: The symbol for an LDR is often drawn with a circle going around it

4. Electricity & Magnetism

YOUR NOTES



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Diagram showing an LDR circuit

4. Electricity & Magnetism

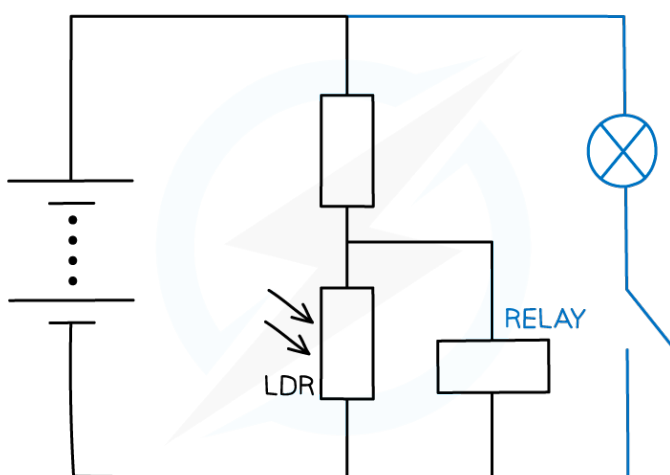
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Sensing Circuits

- Thermistors (or LDRs) may be used to create heat (or light) sensitive circuits, which perform certain actions when the temperature (or light) reaches a certain level



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LDRs can be used in circuits where detecting changes in light intensity are necessary (such as in street lamps)

- In the above circuit, an LDR is used, along with a relay, to control a bulb
 - When it gets darker, the resistance of the LDR increases
 - This causes the potential difference across the LDR to increase
 - When this potential difference increases the current in the coil (part of the relay) will increase
 - When the current gets large enough the coil will attract the switch, closing it and allowing a current to pass through the right-hand side of the circuit
- Temperature controlled circuits may be created by replacing the LDR with a thermistor, and replacing the LDR with a suitable component such as a buzzer or heater

4. Electricity & Magnetism

YOUR NOTES



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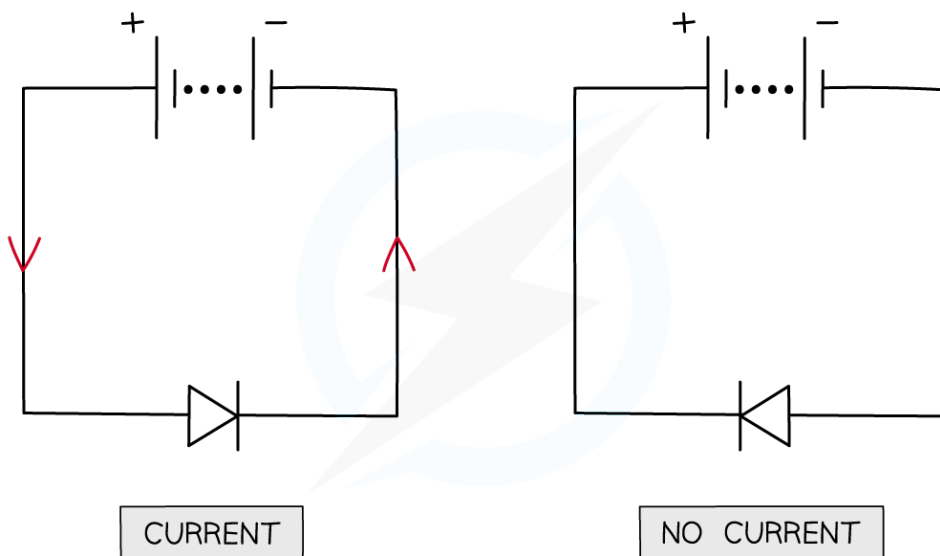
Diode Action



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A diode is a component which only allows a current when the potential difference is in the direction of the arrow

- A diode is a component which only allows charge to flow in one direction through it
- If a power source is connected back to front then there will be no current



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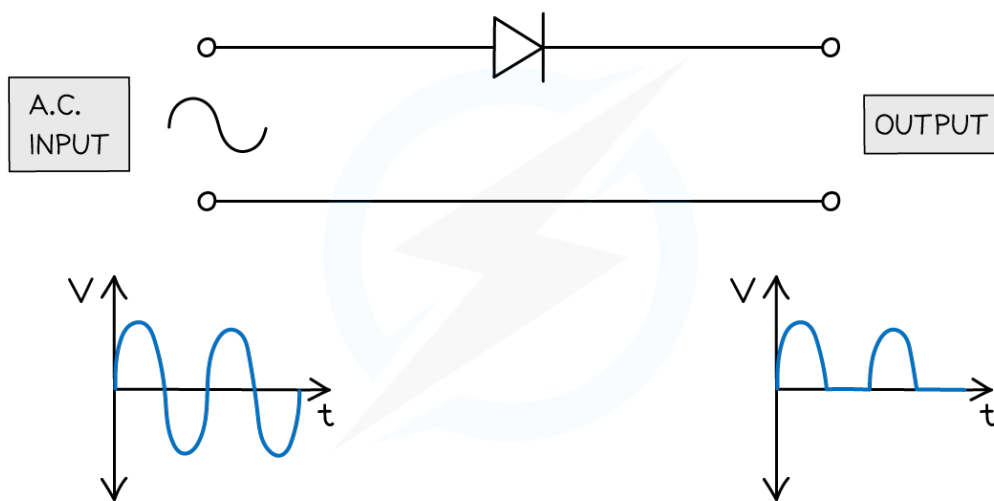
In order to have a current, the diode must point around the circuit from positive to negative

4. Electricity & Magnetism

YOUR NOTES



- If a diode is connected to an a.c. (alternating current) power supply, it will only allow a current half of the time
(This is called rectification)

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A diode can be used to rectify an alternating current

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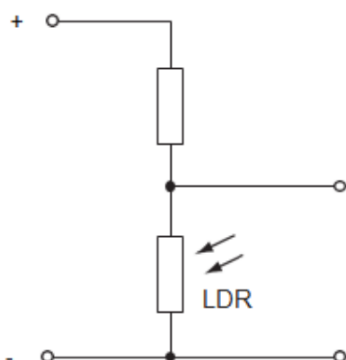
YOUR NOTES



? Exam Question: Medium

A homeowner buys a security light that switches on automatically when it gets dark.

The security light contains the circuit, shown below.



When it gets dark, what happens to the resistance of the LDR, and what happens to the potential difference across the LDR?

	resistance of LDR	p.d. across LDR
A	increases	increases
B	decreases	increases
C	increases	decreases
D	decreases	decreases

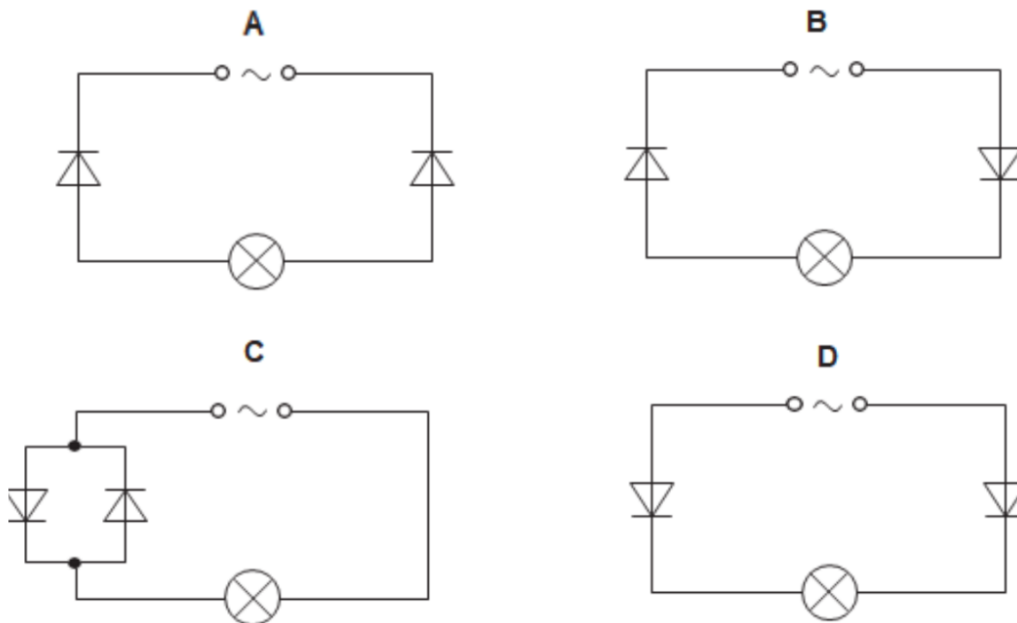
4. Electricity & Magnetism

YOUR NOTES



? Exam Question: Hard

A student sets up four circuits containing diodes. These circuits are shown below.



Which of the circuits will produce a **rectified** current through the bulb?

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4. Electricity & Magnetism

YOUR NOTES



4.4 DIGITAL ELECTRONICS

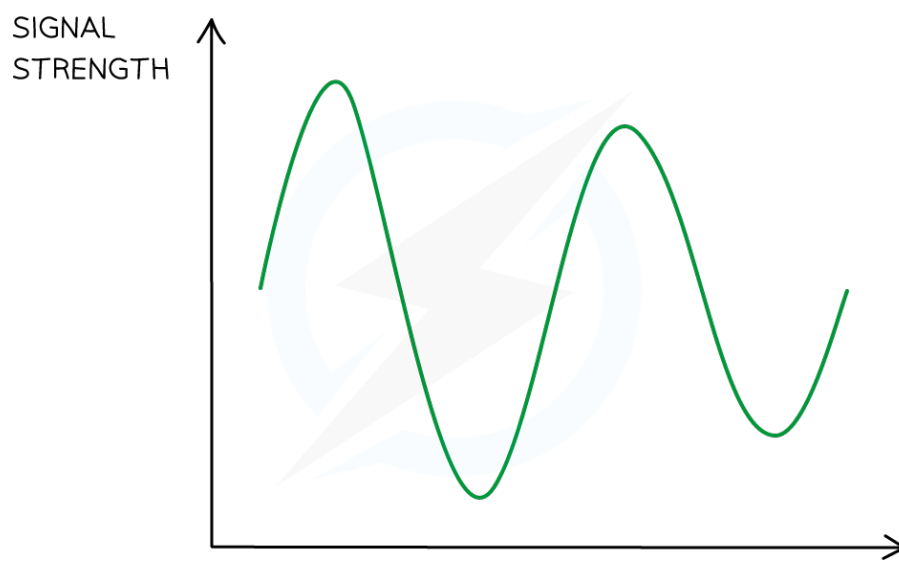
4.4.1 DIGITAL ELECTRONICS



Extended Only

Analogue Signals vs Digital Signals

- There are two types of signals:
 - Analogue
 - Digital
- **An analogue signal is continuously varying** and can take any value



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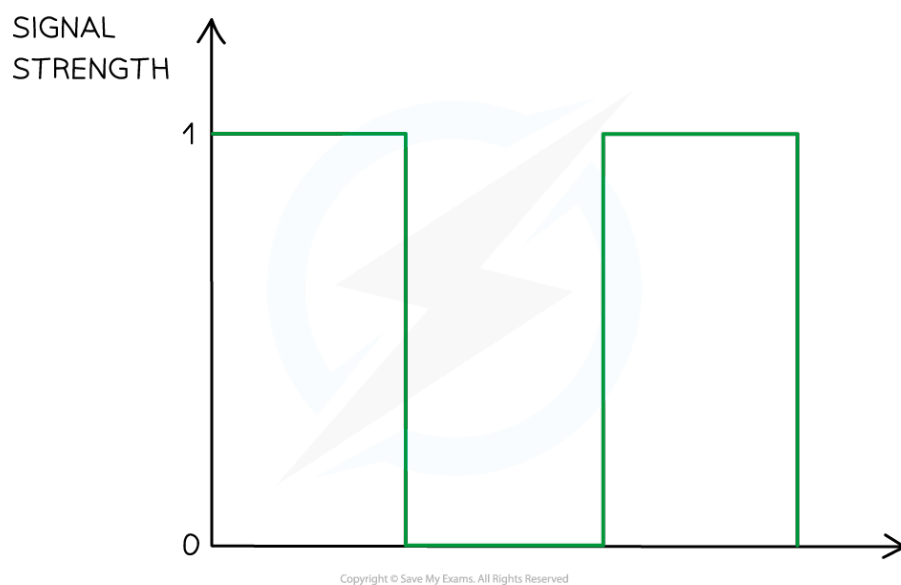
An analogue signal is continuously varying, taking any value

4. Electricity & Magnetism

YOUR NOTES



- A digital signal can only take one of two (discrete) states – commonly referred to as 1s and 0s (or **highs** and **lows**, or **ons** and **offs**)



A digital signal can only take one of two values - 0 or 1

4. Electricity & Magnetism

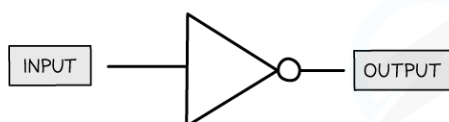
YOUR NOTES



+ Extended Only

Logic Gates

- A logic gate is a digital electronic component that gives a particular output, depending on the input(s) to it
- A **NOT** gate:
 - Has a single input and one output.
 - Will output a **1** if the input is **NOT 1**

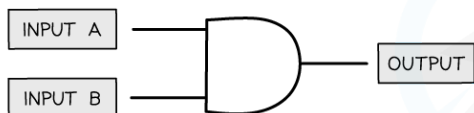


INPUT	OUTPUT
1	0
0	1

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Outputs a 1 if the Input is NOT 1

- An **AND** gate:
 - Has two inputs and one output
 - Will output a **1** if both the first input **AND** the second input are **1**



INPUT A	INPUT B	OUTPUT
0	0	0
0	1	0
1	0	0
1	1	1

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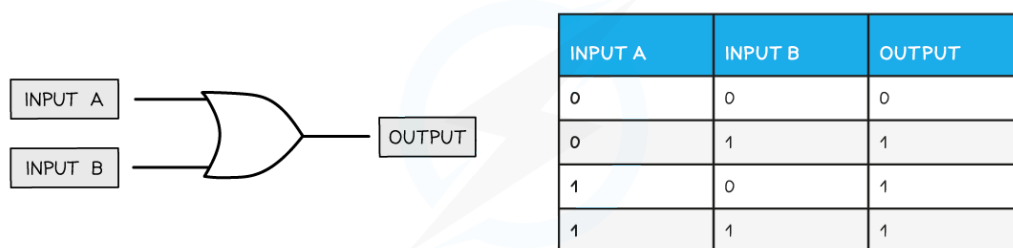
Outputs a 1 if both A AND B are 1

4. Electricity & Magnetism

YOUR NOTES



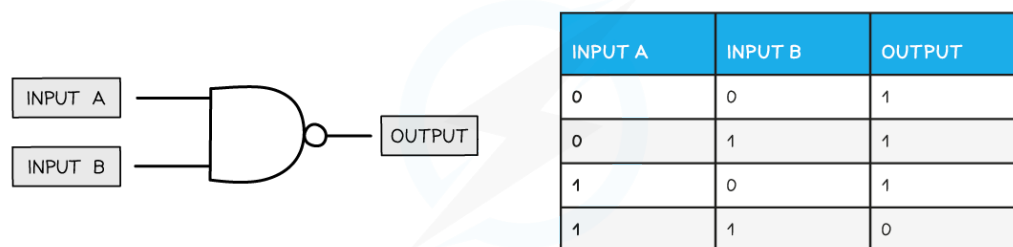
- An **OR** gate:
 - Has two inputs and one output.
 - Will output a **1** if either the first input **OR** the second input is **1**



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Outputs a 1 if either A OR B are 1

- A **NAND** gate:
 - Has two inputs and one output
 - Will output a **1** so long as the first input and the second input are not both **1** (**NOT AND**)



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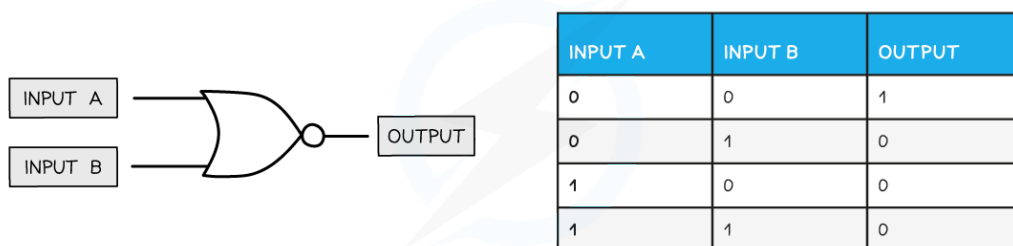
NAND is short for NOT AND - the opposite of an AND gate

4. Electricity & Magnetism

YOUR NOTES



- A **NOR** gate:
 - Has two inputs and one output
 - Will output a **1** if neither the first input **NOR** the second input is **1**



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NOR is short for NOT OR - the opposite of an OR gate

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YOUR NOTES



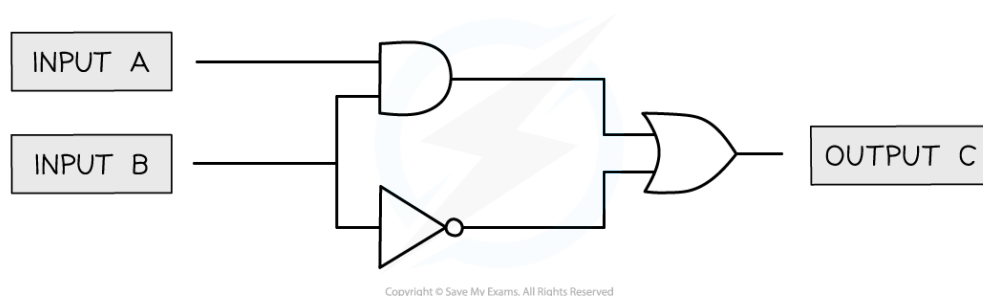
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Truth Tables

- Given a simple circuit involving several logic gates, you should be able to construct a truth table showing how the outputs depend upon the inputs to the circuit
- To solve problems like this, tackle them one step at a time, figuring out how the state of the wires in the middle depends upon the input to the circuit

Example:

- Consider the following circuit, consisting of NOT, AND and OR gates:



A digital circuit, consisting of three different logic gates

- There are two inputs to the circuit, and so there are four different possible inputs:

INPUT A	INPUT B	X	Y	OUTPUT C
0	0			
0	1			
1	0			
1	1			

4. Electricity & Magnetism

YOUR NOTES



(Note how we are also including the states of wires X and Y in our truth table)

- Start by considering how the state of wire X depends on the inputs to the AND gate:

INPUT A	INPUT B	X	Y	OUTPUT C
0	0	0		
0	1	0		
1	0	0		
1	1	1		

- Next let's consider how the state of wire Y depends upon the input to the NOT gate (input B):

INPUT A	INPUT B	X	Y	OUTPUT C
0	0	0	1	
0	1	0	0	
1	0	0	1	
1	1	1	0	

- Finally, we can complete the truth table by considering the inputs to the OR gate (X and Y):

4. Electricity & Magnetism

YOUR NOTES

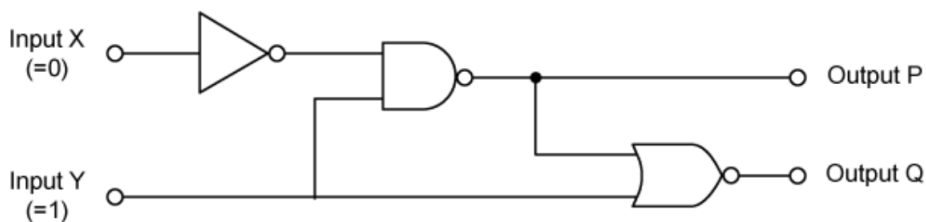


INPUT A	INPUT B	X	Y	OUTPUT C
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1



Exam Question: Medium

The image below shows a set of logic gates.



Input **X** is at a logic state 0 (low) and input **Y** is at a logic state 1 (high).

What are the logic states at output **Q** and **P**?

	output P	output Q
A	1	0
B	1	1
C	0	0
D	0	1

4. Electricity & Magnetism

YOUR NOTES



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4. Electricity & Magnetism

YOUR NOTES



4.5 DANGERS OF ELECTRICITY

4.5.1 HAZARDS, FUSES & EARTHING

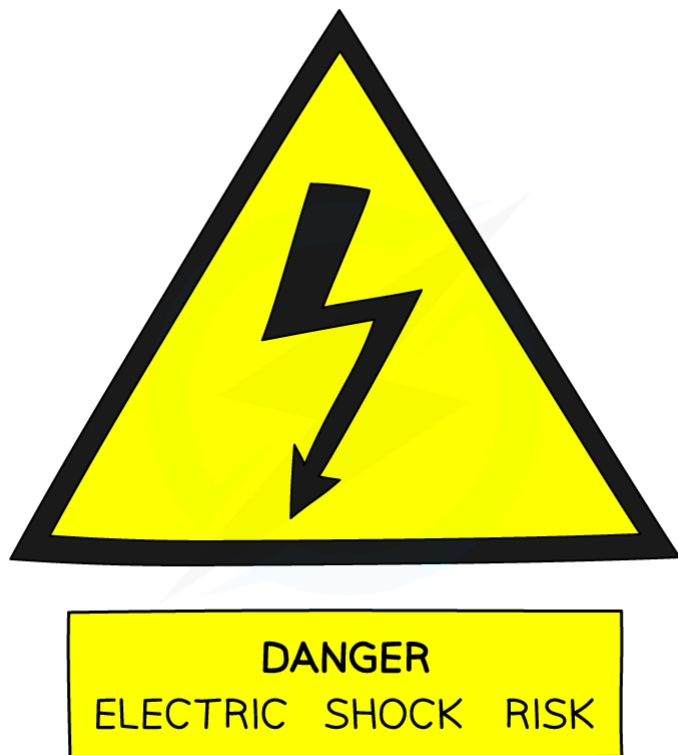
4. Electricity & Magnetism

YOUR NOTES



Hazards of Electricity

- Mains electricity is potentially lethal – potential differences as small as 50 volts can pose a serious hazard to individuals

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Signs, like the above, warn of the risk of electrocution

- Common hazards include:
 - **Damaged Insulation** – if someone touches an exposed piece of wire, they could be subjected to a lethal shock
 - **Overheating of cables** – Passing too much current through too small a wire (or leaving a long length of wire tightly coiled) can lead to the wire overheating. This could cause a fire or melt the insulations, exposing live wires
 - **Damp conditions** – If moisture comes into contact with live wires, the moisture could conduct electricity either causing a short circuit within a device (which could cause a fire) or posing an electrocution risk

4. Electricity & Magnetism

YOUR NOTES



Fuses

- A fuse is a safety device designed to **cut off the flow of electricity to an appliance if the current becomes too large** (due to a fault or a surge)

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The circuit symbol for a fuse - take care not to confuse this with a resistor

- Fuses usually consist of a glass cylinder which contains a thin metal wire.
- If the current in the wire becomes too large:
 - The wire heats up and melts
 - This causes the wire to break, breaking the circuit and stopping the current
- Fuses come in a variety of sizes** (typically 3A, 5A and 13A) - in order to select the right fuse for the job, you need to know how much current an appliance needs
- If you know the power of the appliance (along with mains voltage), the current can be calculated using the equation:

$$\text{CURRENT} = \frac{\text{POWER}}{\text{VOLTAGE}}$$

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- The fuse should always have a current rating that is higher than the current needed by the appliance, without being too high - **always choose the next size up**

Example:

- Suppose an appliance uses 3.1 amps
 - A 3 amp use would be too small - the fuse would blow as soon as the appliance was switched on
 - A 13 amp fuse would be too large - it would allow an extra 10 amps to pass through the appliance before it finally blew
 - A 5 amp fuse would be an appropriate choice, as it is the next size up

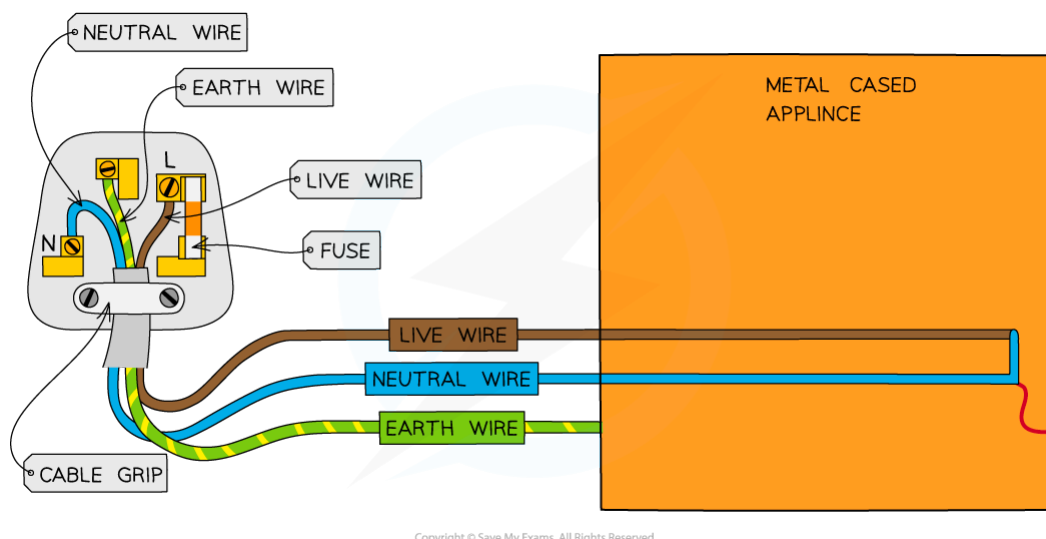
4. Electricity & Magnetism

YOUR NOTES



Earthing

- Many electrical appliances have metal cases
- This poses a potential safety hazard:
 - If a live wire (inside the appliance) came into contact with the case, the case would become electrified and anyone who touched it would risk electrocution
- The earth wire is an additional safety wire that can reduce this risk



A diagram showing the three wires going to a mains powered appliance: live, neutral and earth

- If this happens:
 - The earth wire provides a **low resistance path to the earth**
 - This causes a **surge of current in the earth wire** and hence also in the live wire
 - The high current through the fuse causes it to **melt and break**
 - **This cuts off the supply of electricity to the appliance**, making it safe

4. Electricity & Magnetism

YOUR NOTES



Exam Question: Medium

The plug for an electric lamp contains a 3 A fuse. The fuse's job is to protect the lamp if too much current flows.

Where is the fuse positioned, and what happens when the fuse does its job?

	where is it positioned?	what does it do?
A	live wire	reduces the current to 3 A
B	neutral wire	breaks the circuit
C	live wire	breaks the circuit
D	neutral wire	reduces the current to 3 A

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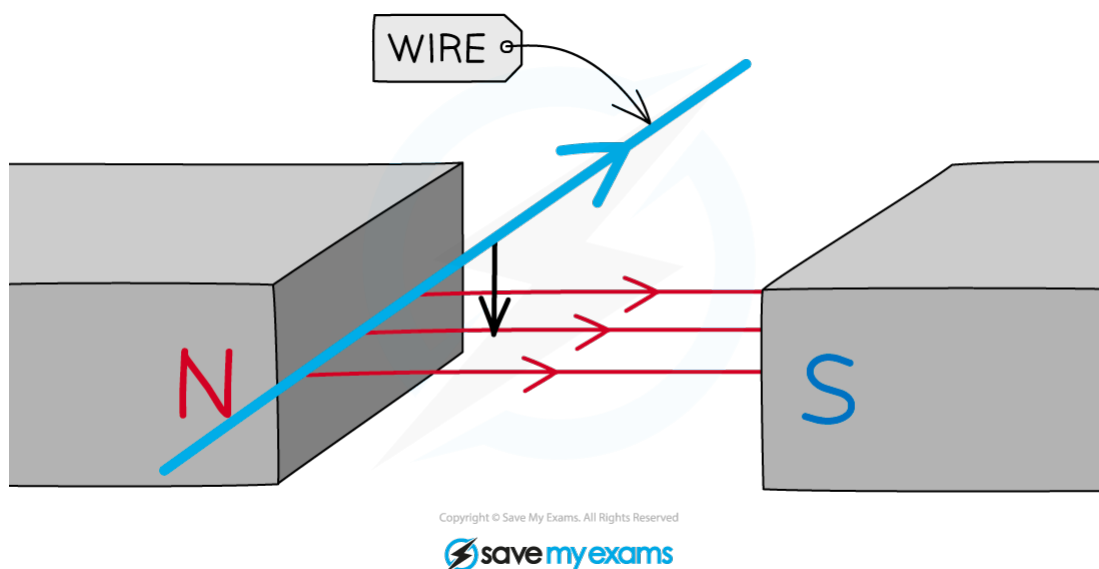


4.6 ELECTROMAGNETIC EFFECTS

4.6.1 ELECTROMAGNETIC INDUCTION

Inducing an EMF in a Conductor

- When a conductor (such as a metal wire) is moved through a magnetic field, **it cuts through the field lines, inducing an EMF in the wire**



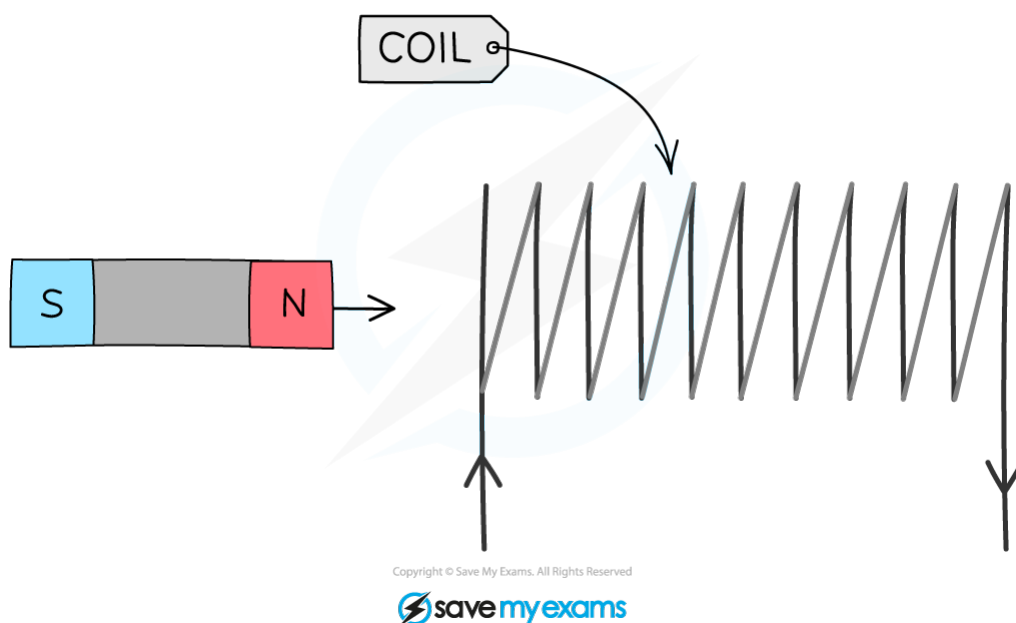
As the wire moves downwards, it cuts through field lines, inducing an EMF in the wire

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- A similar effect occurs if a magnet is pushed into, or taken out of a coil:
 - As the magnet moved through the coil, the field lines cut through the turns on the coil
 - This induces an emf in the coil



When the magnet enters the coil, the field lines cut through the turns, inducing an EMF

- More generally, whenever the magnetic field passing through a loop of wire changes, an EMF is induced

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YOUR NOTES



Factors Affecting EM Induction

- The size of the induced EMF is proportional to the rate at which the field lines are cut:
 - If the field lines are cut at a faster rate, the EMF will increase
- This can be achieved by:
 - **Moving the wire (or magnet) faster**
 - **Using a stronger magnet** (increasing the number of field lines)
 - **Adding more turns to a coil** (assuming a coil is being used, and not just a wire)



Exam Tip

When discussing factors affecting EM Induction:

- Make sure you state:

“Add more turns to the coil”

- And not just:

“Add more coils”

(This second one means something slightly different)

- Likewise, when referring to the magnet, use the phrase:

“A stronger magnet”

- And not:

“A bigger magnet”

(Large magnets are not necessarily stronger)

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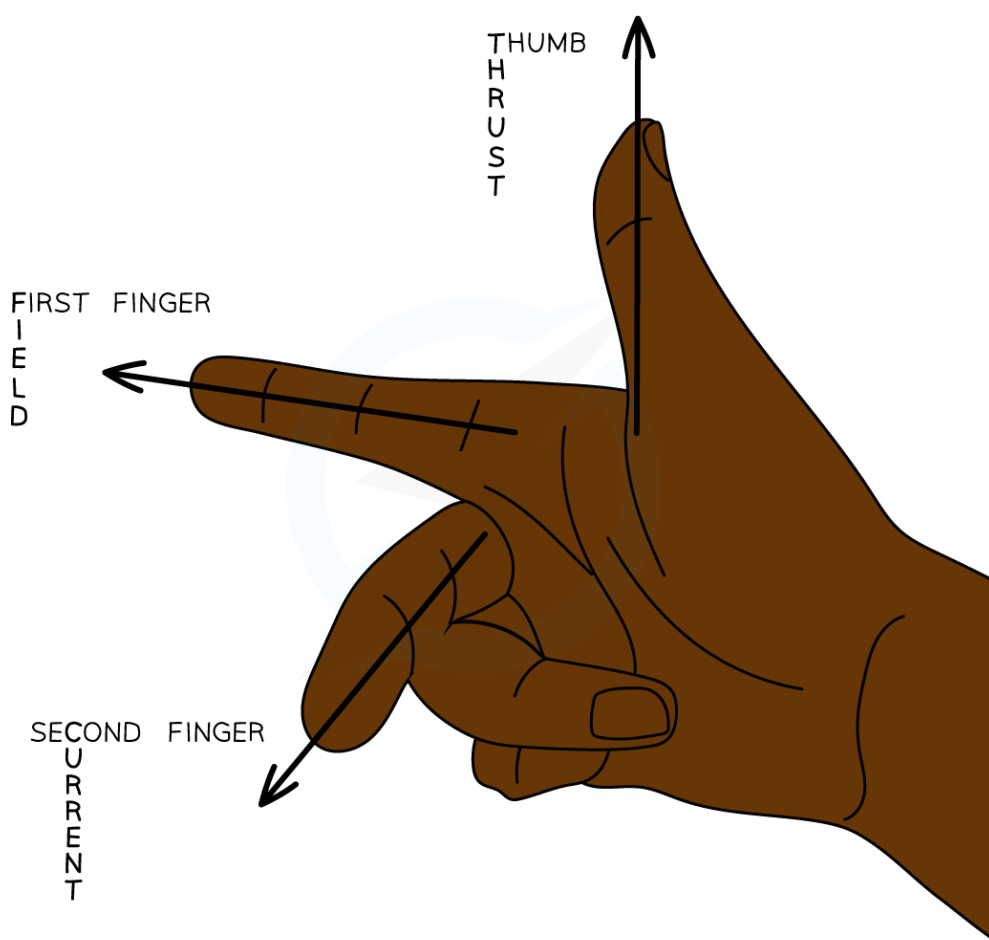
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The Right-Hand Rule

- When moving a wire through a magnetic field, the direction of the induced EMF can be worked out by using the Right-Hand Dynamo rule:



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The Right-Hand Dynamo rule can be used to deduce the direction of the induced EMF

- To use the rule:
 - Start by pointing the **first finger** (on your right hand) in the direction of the **field** (**F**irst **F**inger **F**ield)

4. Electricity & Magnetism

YOUR NOTES



- Next, rotate your hand so that the thumb point in the direction that the wire is moving in
(ThMb Motion)
- Your SeCond finger will now be pointing in the direction of the current (or, strictly speaking, the EMF)
(SeCond Current)
- **The direction of the induced EMF always opposes the change that produces it**
- This means that any magnetic field created by the EMF will act so that it tried to stop the wire or magnet from moving

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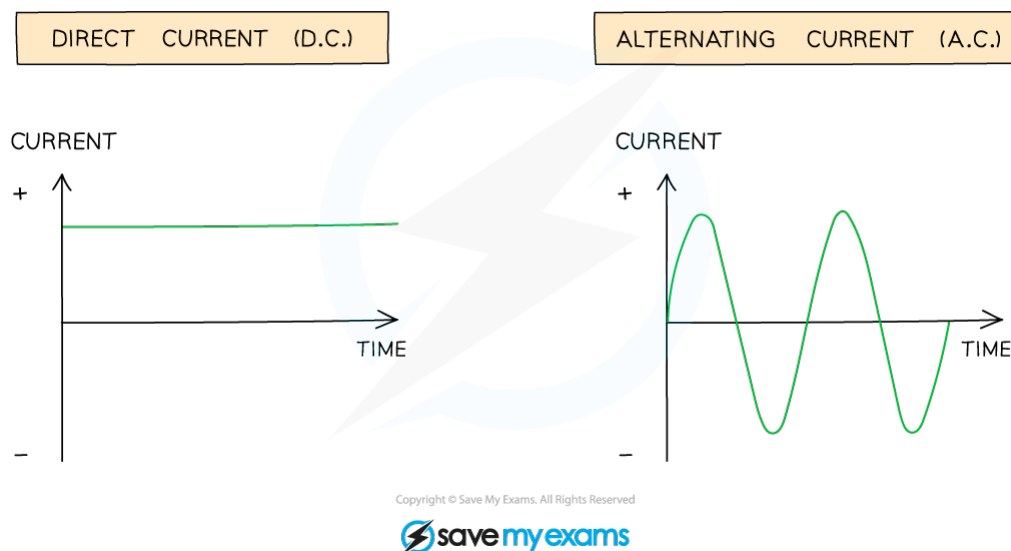
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4.6.2 A.C. & D.C.

Alternating vs Direct Current

- An **alternating current (a.c.)** is one that **continuously changes its direction**, going back and forth around a circuit
- A **direct current (d.c.)** is one that is **steady, constantly going the same way around a circuit**, from positive to negative



Two graphs showing the variation of current with time for alternating current and direct current

- Electric cells (batteries) produce direct current (d.c.)
- Mains electricity is an alternating current

4. Electricity & Magnetism

YOUR NOTES



Exam Tip

If asked to explain the difference between alternating and direct current, sketch the graphs shown above: a well-sketched (and labelled) graph can earn you full marks.

4. Electricity & Magnetism

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4.6.3 A.C. GENERATOR

4. Electricity & Magnetism

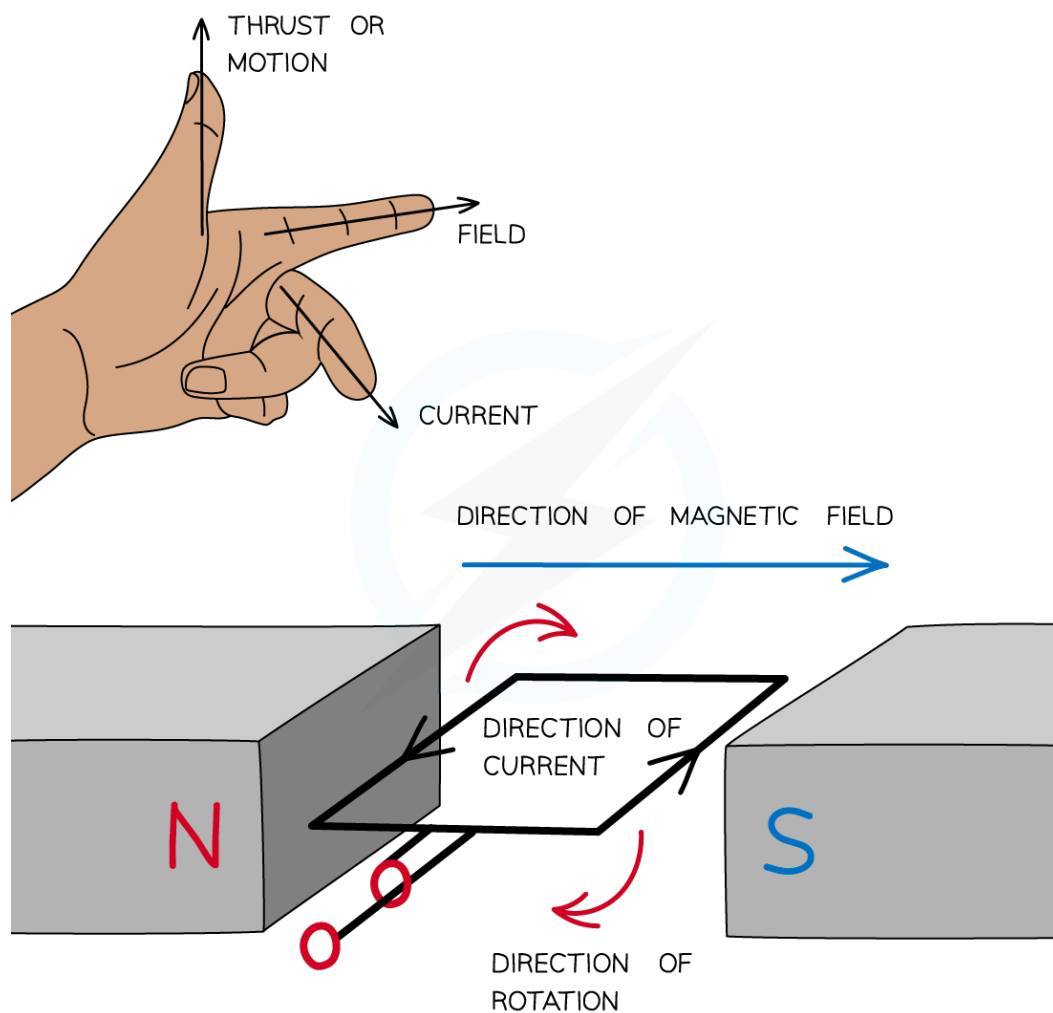
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A.C. Generator: Basics

- A generator looks very similar to a motor, but instead of connecting it to a power supply, the coil is spun by some mechanical process which then produces electricity



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When a coil is spun in a magnetic field, a voltage is induced between the ends of the coil

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- As the coil rotates, it **cuts through the field lines**
- This **induces an EMF** between the end of the coil (which could then create a current)
- The size of this EMF could be increased by:
 - Turning the coil faster
 - Increasing the strength of the magnetic field
 - Adding more turns to the coil
- Slip rings, attached to the ends of the coil, transfer the current to metal brushes whilst allowing the coil to rotate freely



Exam Tip

When a conductor **cuts through field lines**, it **induces an EMF**.

Motors and generators look very similar, but they do very different things.

When tackling a question on either of them, make sure you are writing about the right one!

Remember to say 'add more **turns** to the coil' and not 'add more coils' – if you give the second one it will be marked wrong.

4. Electricity & Magnetism

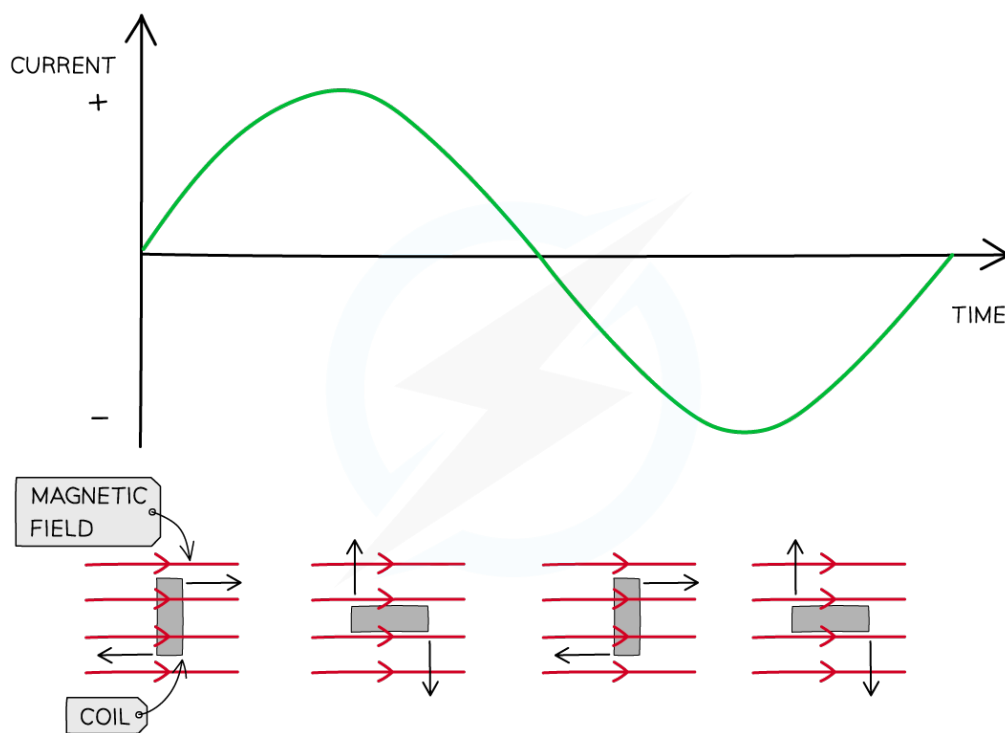
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The Output

- The A.C. generator creates an alternating current, varying in size and direction as the coil rotates
 - The induced EMF is greatest when the coil is horizontal, as in this position it cuts through the field at the fastest rate
 - The EMF is smallest when the coil is vertical, as in this position it will not be cutting through field lines



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Diagram showing how the current varies with the position of the coil

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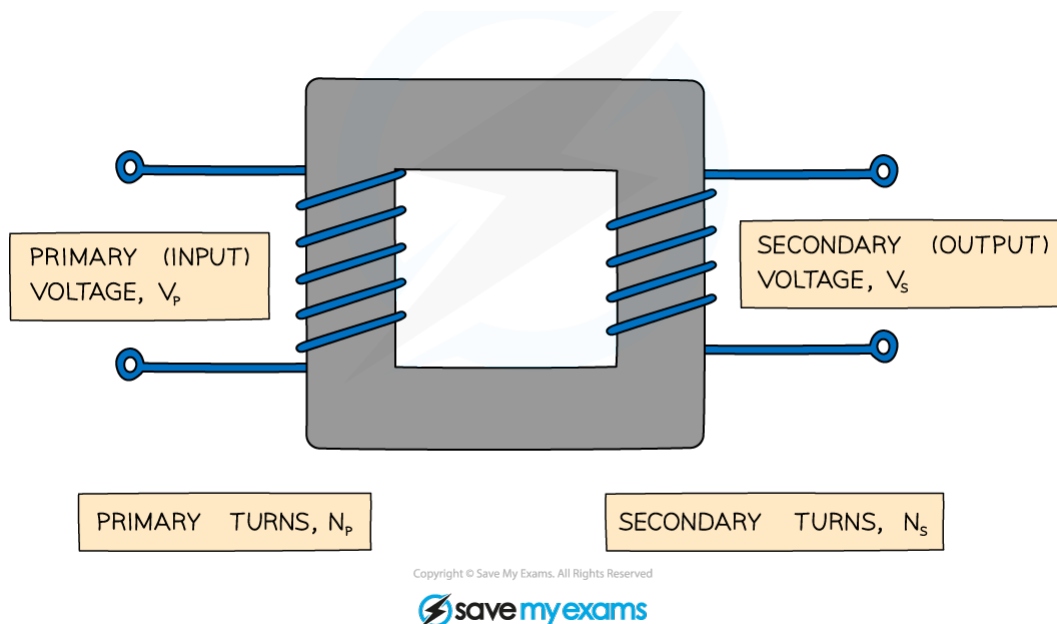


4.6.4 TRANSFORMERS

What is a Transformer?

- A transformer is an electrical device that can be used to increase or decrease the voltage of an alternating current

(Transformers only work with a.c.)



A transformer consists of two coils of wire wrapped around a soft iron core

- A **step-up** transformer **increases the voltage** of a power source
- A step-up transformer has more turns on the secondary coil than on the primary coil
- A **step-down** transformer **decreases the voltage** of a power source
- A step-down transformer has fewer turns on the secondary coil than on the primary coil

4. Electricity & Magnetism

YOUR NOTES



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How a Transformer Works

- When an alternating current is supplied to the primary coil, a changing magnetic field is produced by the primary coil
- This field passes through the soft iron core and through the secondary coil
- The changing field in the secondary coil induces an EMF
- This EMF is also alternating and has the same frequency as the original current

The Transformer Equation

- The output voltage of a transformer depends on the number of turns on each of the coils as well as the input voltage:

$$\frac{\text{input (primary) voltage}}{\text{output (secondary) voltage}} = \frac{\text{primary turns}}{\text{secondary turns}}$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

(The ratio of the voltages is equal to the ratio of the number of turns)

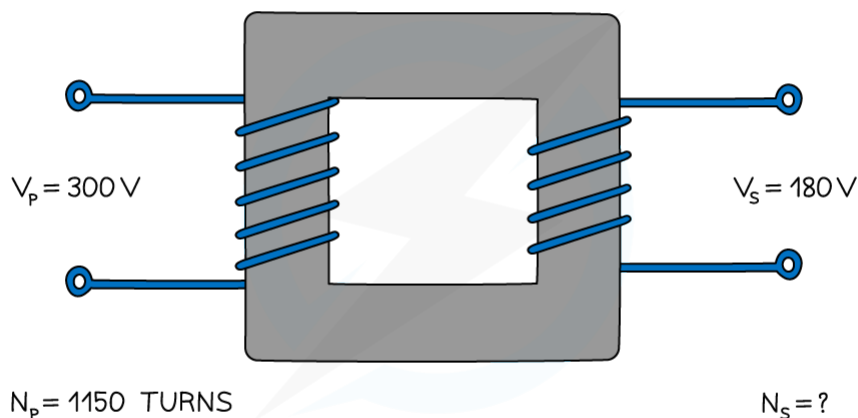
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*input power = output power*

$$I_P V_P = I_S V_S$$

WORKED EXAMPLE: IF A STEP-DOWN TRANSFORMER TURNS A PRIMARY VOLTAGE OF 300V DOWN TO A SECONDARY VOLTAGE OF 180V, WHAT IS THE NUMBER OF TURNS NEEDED IN THE SECONDARY COIL IF THE PRIMARY COIL CONTAINS 1150 TURNS OF WIRE ?



$$\frac{V_p}{V_s} = \frac{N_p}{N_s} \quad \longrightarrow \quad \frac{300}{180} = \frac{1150}{N_s} \quad \longrightarrow \quad \frac{180}{300} \times 1150 = N_s$$

$$N_s = 690 \text{ TURNS}$$

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Transformer calculation worked example

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Power

- Although transformers can increase the voltage of a power source, due to the law of conservation of energy, they cannot increase the power output (Increasing the voltage decreases the current, and vice-versa)
- **If a transformer is 100% efficient:**

input power = output power

$$I_P V_P = I_S V_S$$

(Where I_P and I_S are the currents in the primary and secondary coils)

- Hence, if the voltage is increased by some factor, the current must decrease by the same factor

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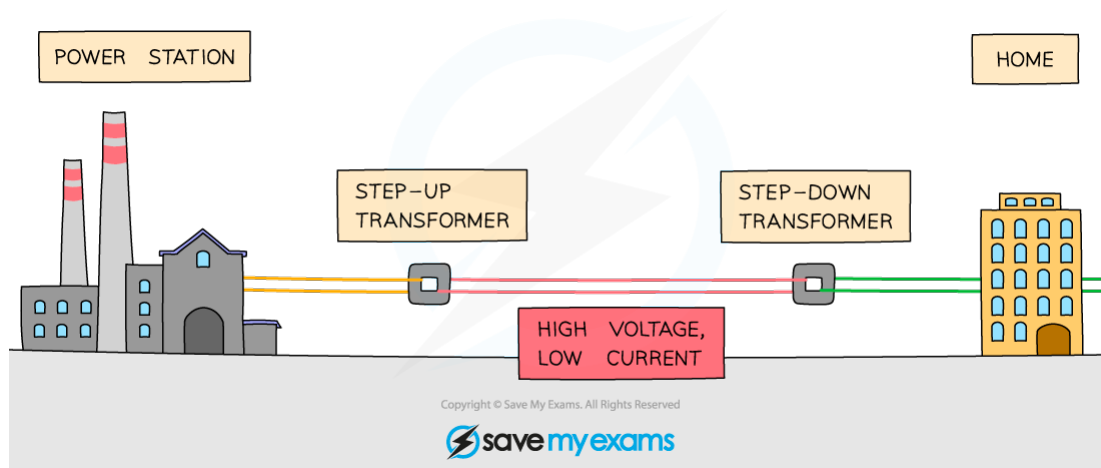
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4.6.5 TRANSMITTING ELECTRICITY

High-Voltage Transmission

- When electricity is transmitted along overhead cables, it is done at high voltages
 - A step-up transformer is used to raise the potential difference (voltage) before transmissions
 - A step-down transformer is then used to step the potential difference back down to normal levels when it reaches its destination



Electricity is transmitted at high voltage, reducing the current and hence power loss in the cables

4. Electricity & Magnetism

YOUR NOTES



Extended Only

How High Voltage Reduces Power Loss

- When electricity is transmitted over large distances, the current in the wires heats them, resulting in energy loss
- By raising the voltage at which the electricity is transmitted at, the same amount of power (energy per second) can be transmitted using a much smaller current ()
- This results in less heat being produced in the wire and hence less energy loss

4. Electricity & Magnetism

YOUR NOTES



4.6.6 THE MAGNETIC EFFECT OF A CURRENT

The Field around a Wire

- When there is a current in a wire, a magnetic field is created around the wire

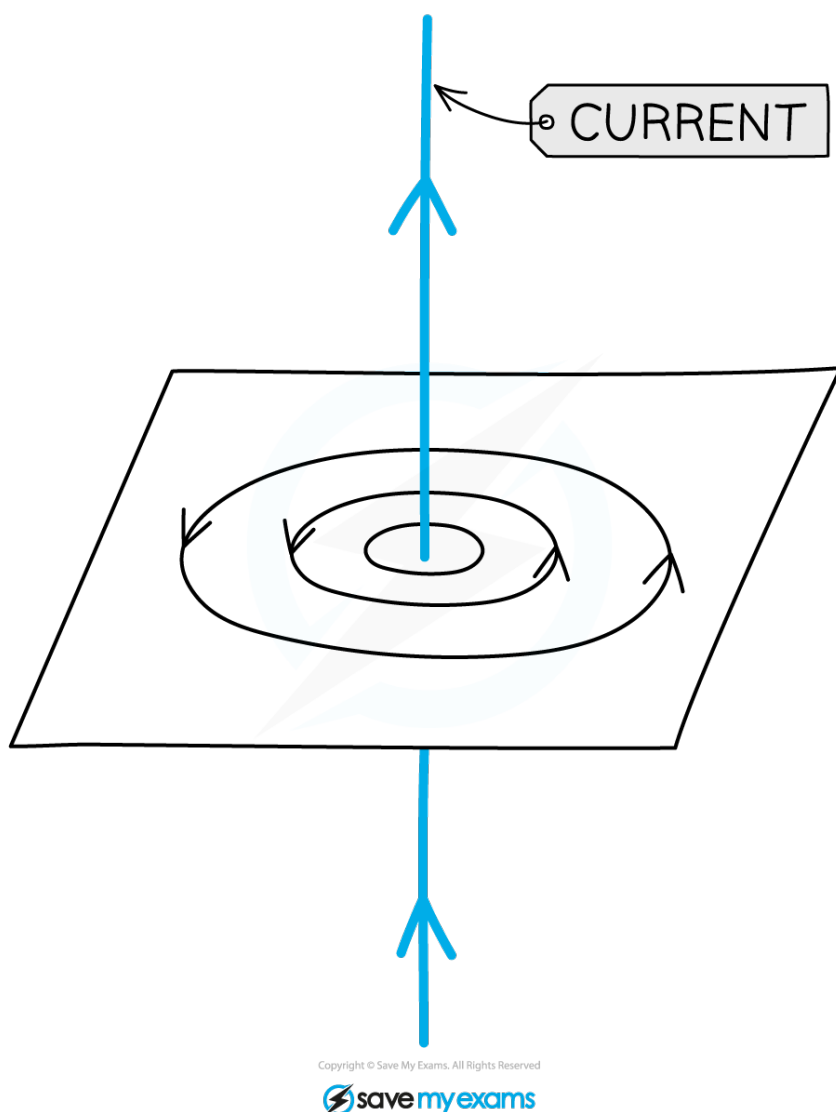


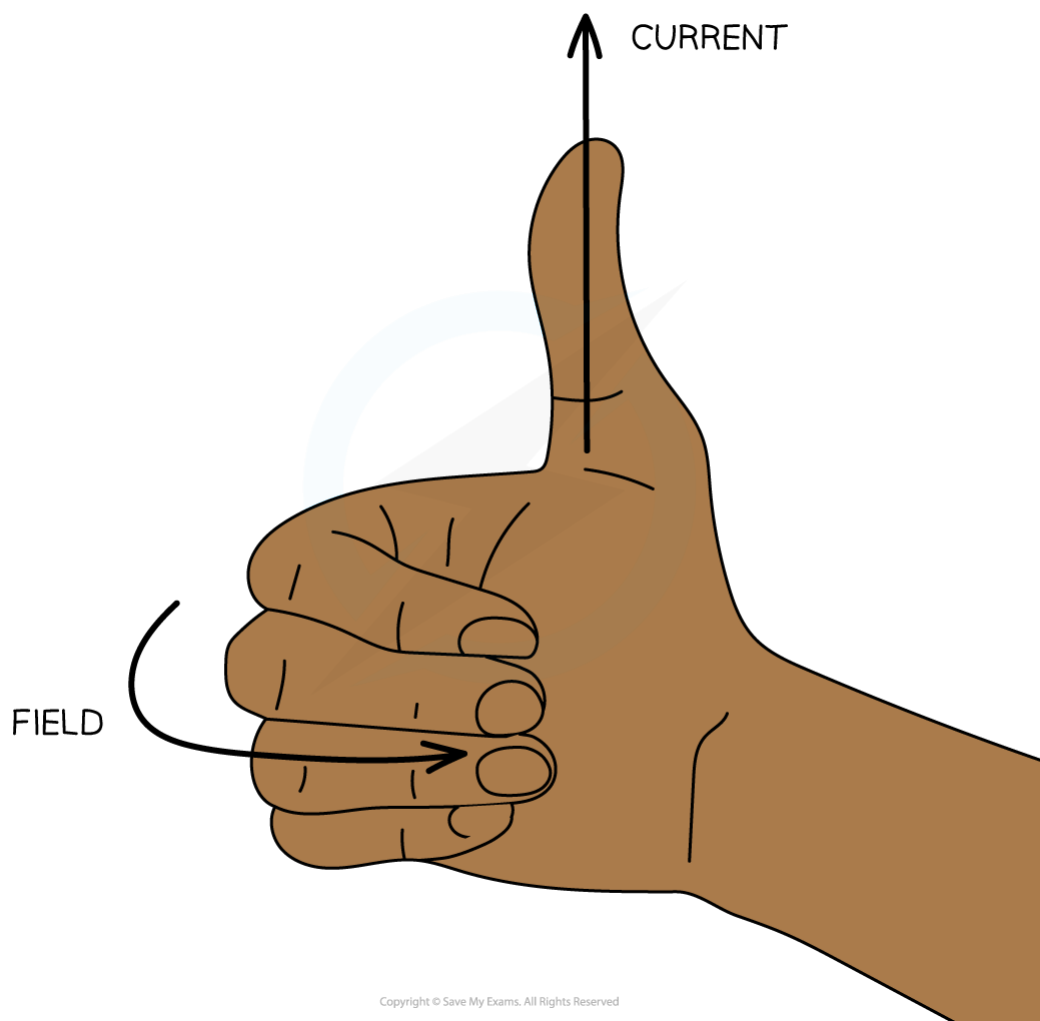
Diagram showing the magnetic fields around a current-carrying wire

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- The magnetic field lines form concentric circles around the wire
- The direction of the field is given by the right-hand grip rule:



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When the thumb is pointing in the direction of the current, the fingers will curl in the direction of the field

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Magnetic Field Strength & Direction

- The direction of a magnetic field is defined as being:
 - The direction of the force on the north pole of a magnet placed at that point
- The strength and direction of the field depend on the size and direction of the current:
 - If the current is increased, the field will get stronger
 - If the direction of the current is changed, the direction of the field will change
- The strength of the magnetic field is related to the distance between the field lines:
As the field lines spread out, the field gets weaker
- The field lines around a wire get further apart the further they are from the wire

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The Field around a Coil

- When a wire is looped into a coil, the magnetic field lines circle around each part of the coil, passing through the centre of it

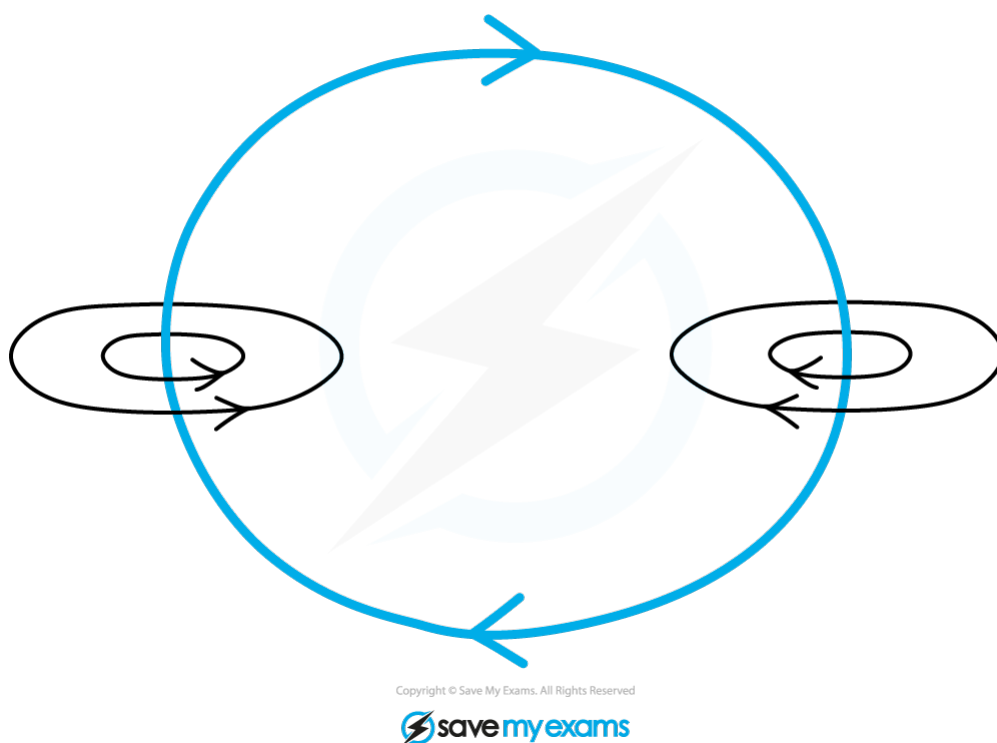


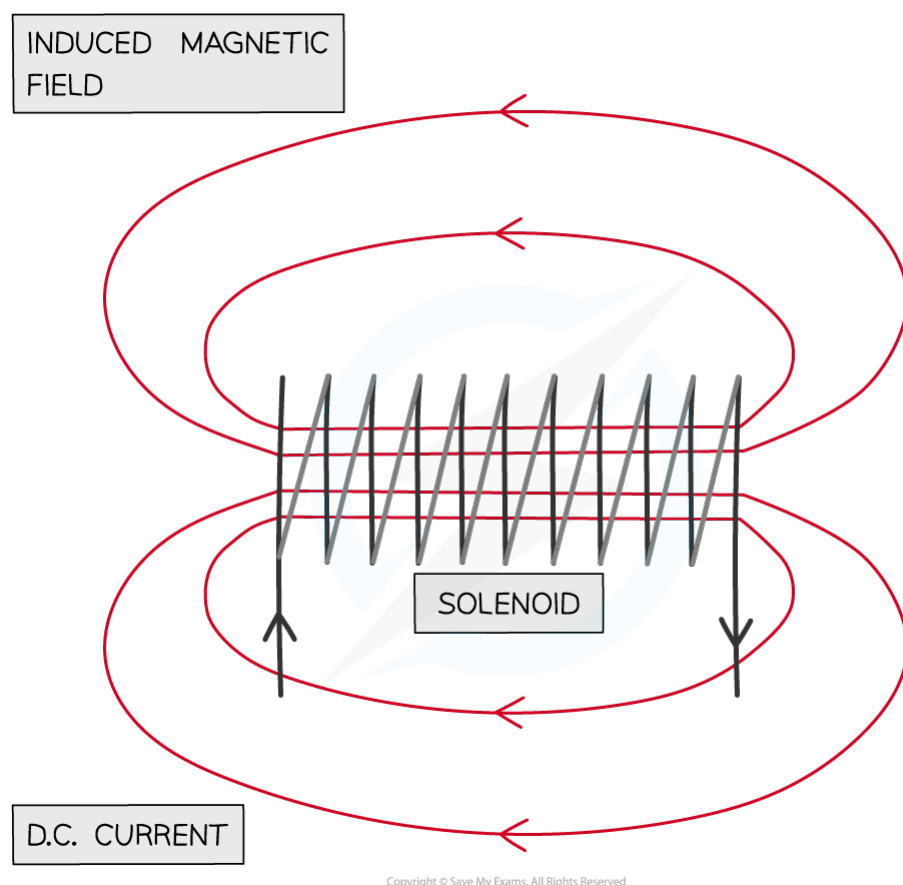
Diagram showing the magnetic field around a flat circular coil

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- **A solenoid is a long coil**
- The magnetic field around a solenoid look the same as the magnetic field around a bar magnet:



The magnetic field around a solenoid (a long coil) is identical to the magnetic field of a bar magnet

- One end of the solenoid behaves like the north pole of a magnet; the other side behaves like the south pole

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YOUR NOTES



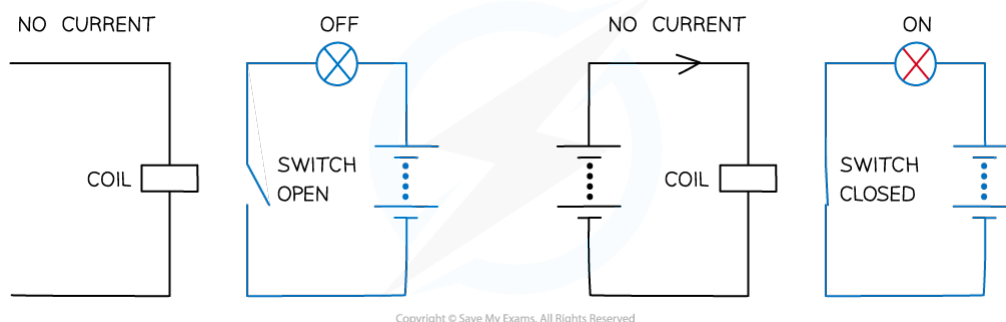
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Strength & Direction within a Solenoid

- Inside the solenoid the field lines straighten up and are very close together - they form a strong uniform field

Solenoid Applications

- A solenoid can be used as an electromagnet by adding a soft iron core (This increases the strength of the magnetic field significantly)
- Electromagnets are used in a wide variety of applications including:
 - Door bells
 - Electronic door locks
- An electromagnet is also used in a relay:



When a current passes through the coil, it attracts the switch, closing it which allows a current in the right-hand part of the circuit

4. Electricity & Magnetism

YOUR NOTES



4.6.7 FORCE ON A CURRENT-CARRYING CONDUCTOR

The Motor Effect

- When a current passes through a wire in a magnetic field, a force is exerted on the wire

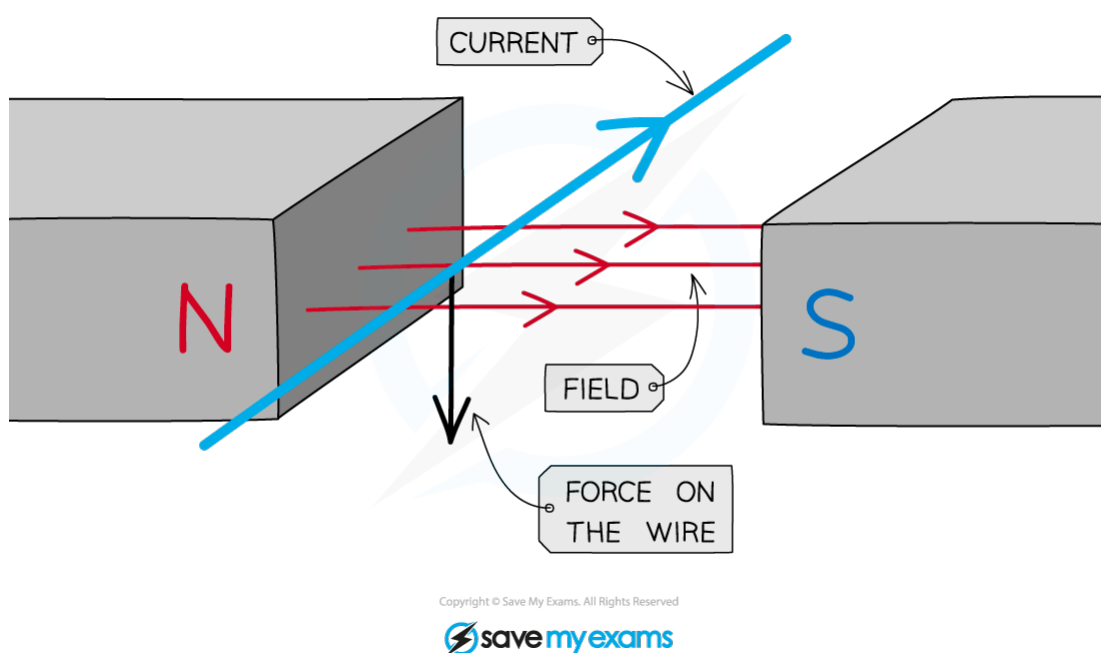


Diagram showing the force acting on a current-carrying wire in a magnetic field

- The direction of this force depends on:
 - The direction of the field
 - The direction of the current
- Reversing either of the above will reverse the direction of the force

4. Electricity & Magnetism

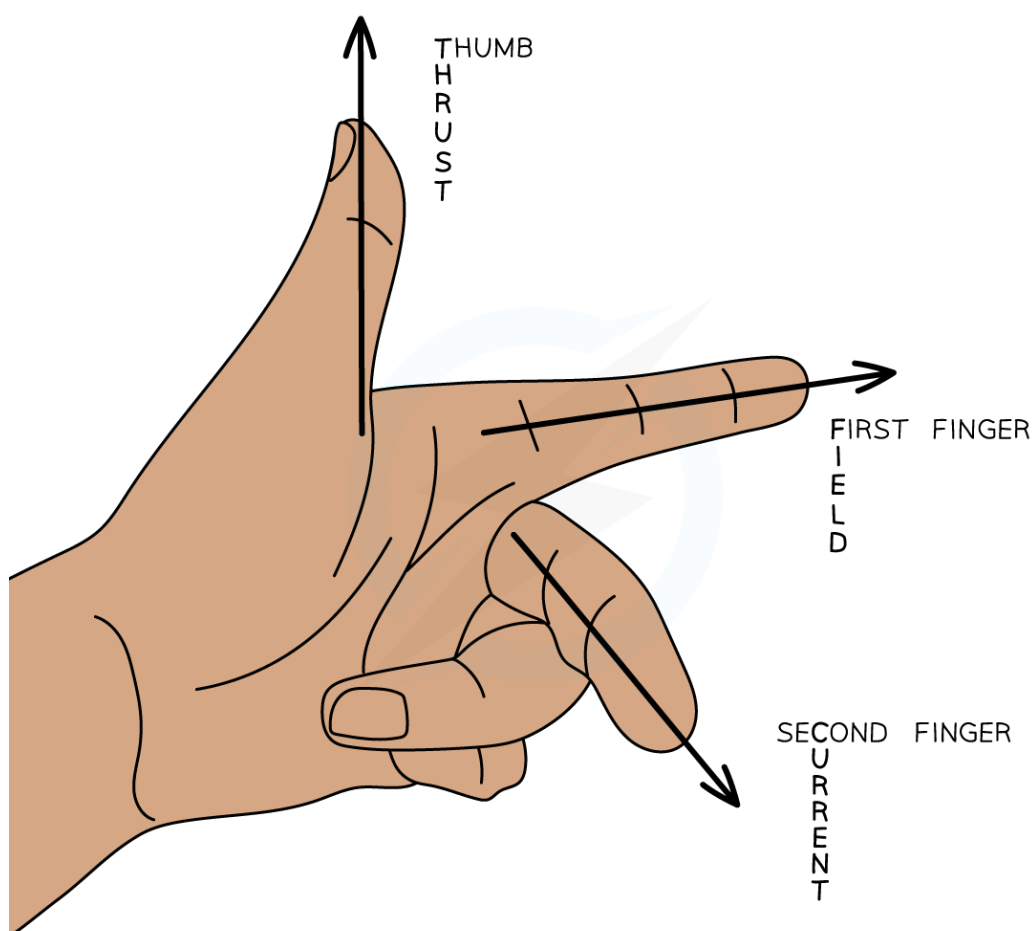
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The Left-Hand Rule

- The force is always directed at 90 degrees to both the field and the current
- The direction of the force on a current-carrying wire can be worked out by using the left-hand rule:

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The left-hand rule can help you figure out the direction of the force on a current-carrying wire

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- Start by pointing your **F**irst **F**inger in the direction of the (magnetic) **F**ield
- Now rotate your hand around the first finger so that the se**C**ond finger points in the direction of the **C**urrent
- The **T**Humb will now be pointing in the direction of the **T**Hrust (the force)

4. Electricity & Magnetism

YOUR NOTES

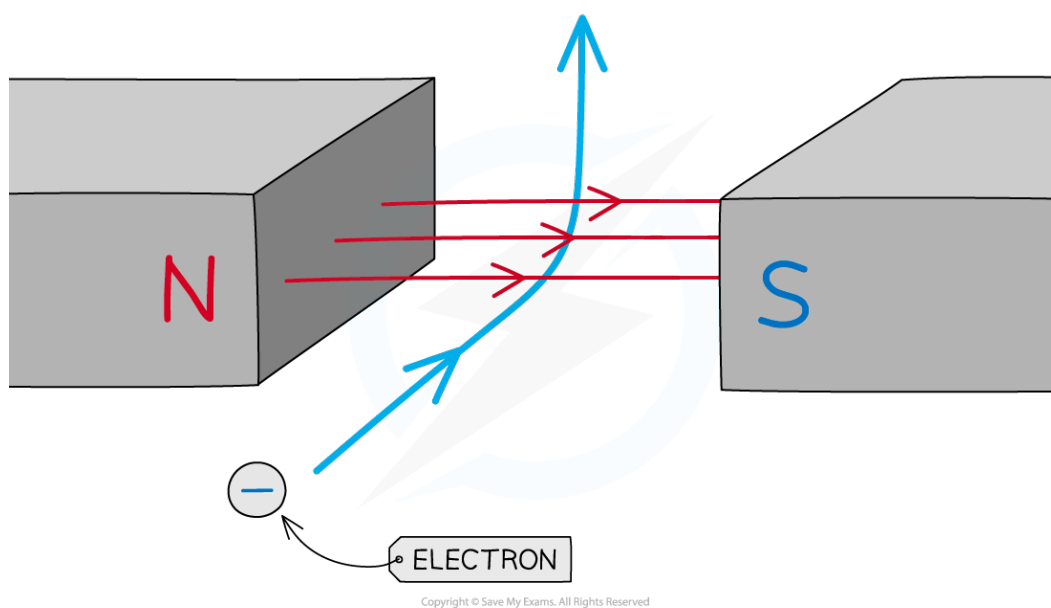


4.6.8 FORCE ON A CHARGED PARTICLE

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How a Magnetic Field Deflects a Charged Particle

- When a charged particle passes through a magnetic field, the field can exert a force on the particle, deflecting it
- This happens because the moving charge forms a current



When a charged particle (such as an electron) enters a magnetic field, it is deflected by the field

4. Electricity & Magnetism

YOUR NOTES



- The **force is always at 90 degrees** to both the direction of travel and the magnetic field lines, and can be worked out by using the left-hand rule
 - **However**, if the particle has a **negative charge** (such as an electron), then the second finger (the current) must point in the opposite direction to the direction of travel
- The deflection of charged particles can be demonstrated either by using a cathode ray tube and a pair of magnets, or by passing a collimated beam of beta particles (high energy electrons) between the poles of a horseshoe magnet

(Note: A cathode ray tube fires electrons at high speed towards a target. Old TV sets contained cathode ray tubes, but you must be careful using these, as holding a magnet to the screen can permanently affect the image)

4. Electricity & Magnetism

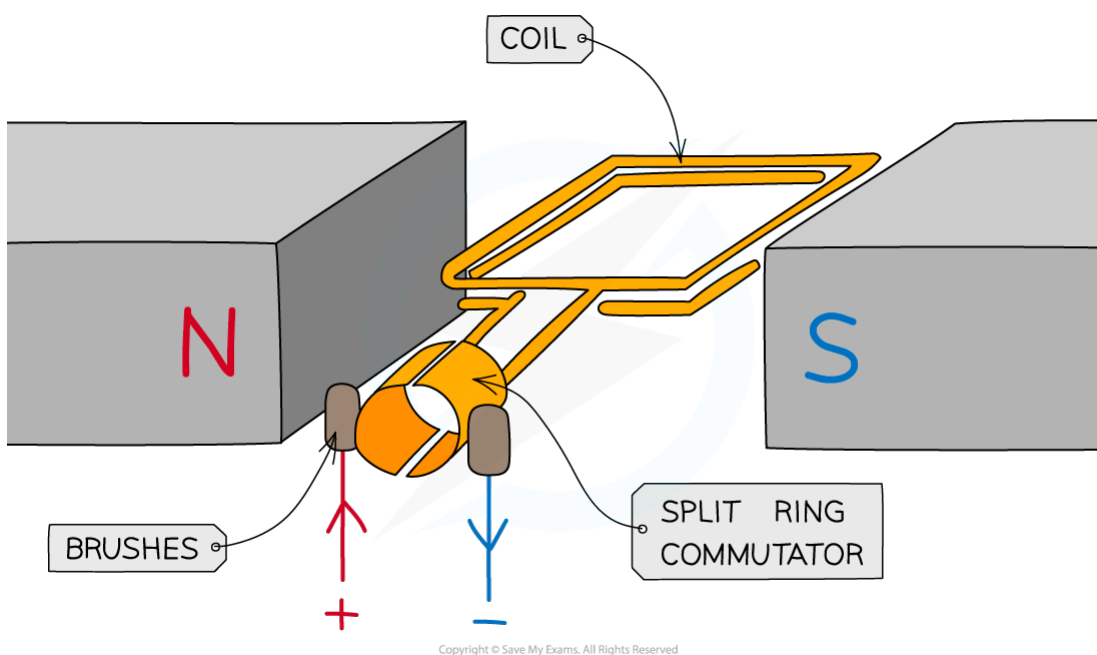
YOUR NOTES



4.6.9 D.C. MOTOR

Simple D.C. Motor: Basics

- The simple d.c. motor consists of a coil in a uniform magnetic field:



The simple d.c. motor

- When there is a current in the coil, the magnets exert a turning effect on the coil, causing it to rotate
- The force supplied by a motor can be increased by:
 - Increasing the current in the coil
 - Increasing the strength of the magnetic field
 - Adding more turns to the coil

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Extended Only

How a Simple Motor Works

- When a current passes through the coil:
 - The current creates a magnetic field around the coil,
 - which interacts with the field of the magnets,
 - exerting a force on the coil,
 - in accordance with the left-hand rule
 - This pushes one side of the coil up and the other side down, causing it to spin
- The commutator:
 - Reverses the direction of the current in the coil every half turn
 - This reverses the direction of the forces, which keeps the coil spinning

4. Electricity & Magnetism

YOUR NOTES

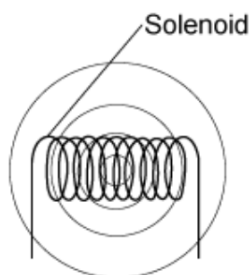


? Exam Question: Easy

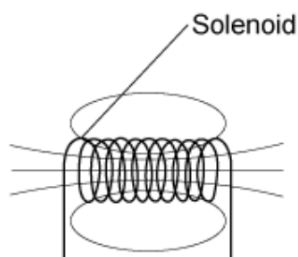
Four students, of varying ability in physics, draw what they believe to be the field pattern of a current-carrying solenoid.

Which of them is correct?

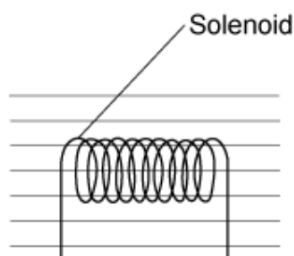
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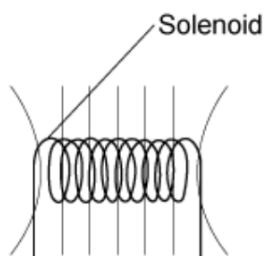
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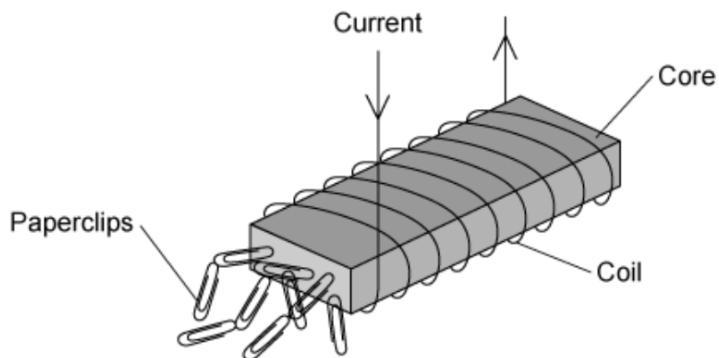
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Exam Question: Hard

The diagram below shows an electromagnet. A student wants to find out how strong the magnetic field is by counting the number of metal pins it can pick up.



The experiment is repeated but the current in the coil is doubled. What would happen to the number of pins attracted to the electromagnet?

- A More pins would be attracted.
- B Some pins would fall off.
- C The number of pins would stay the same.
- D No pins would be attracted.

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